

A Bloomberg Professional Services Offering

BTST <GO> Basic Backtesting Help Page

Enter BTST <GO>, then press <Help>

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Definitions

For terms in *blue* throughout this document, see [Definitions](#) on p.52.

BTST <GO>

BTST allows you to run a pre-configured multi-strategy analysis for the currently loaded security or group of securities, so you can assess the effectiveness of a variety of trading strategies over the date range of the analysis. You can run the BTST test for a historical or intraday time period, set parameters for the trades generated by the test, and determine the way commissions generated by the test are calculated. The function can be used to test strategies for trading all asset classes. The BTST analysis appears as a worksheet on the Backtesting (BT) function Multi-Strategy Worksheets/Sample Worksheets tab and can be run any time.

Backtesting & Optimization | BT <GO>

BT <GO> lets you backtest trading strategies, optimize strategy parameters, and visualize test results with historical or intraday data from any time period, so you can assess the effectiveness of entry/exit rules that you define.

You can backtest a single strategy or multiple strategies at once across all asset classes for any number of securities.

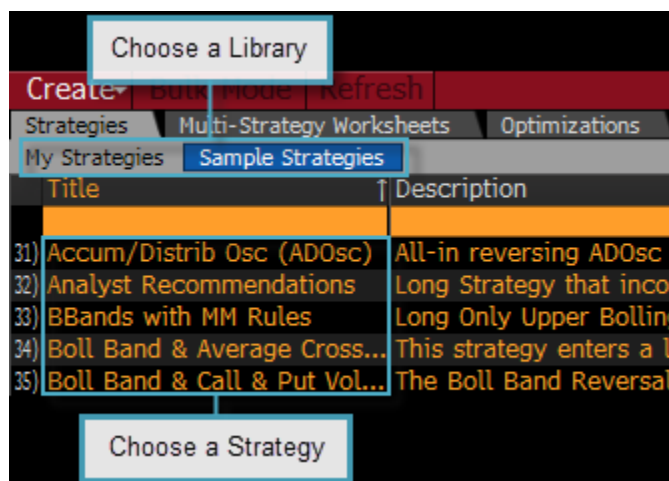


Create • Backtesting & Optimization	
Strategies Multi-Strategy Worksheets Optimizations Multi-Security Analyses	
My Strategies Sample Strategies	
Title	Description
10 52 Week Slope	Buy when 52 week Moving Linear Regression Slope of price is positive.
13 Accum/Distrib Osc	All-in reversing Accumulation/Distribution Oscillator Strategy goes long when the ADO li...
13 Analyst Recommendations	Long Strategy that incorporates Buy & Sell Recommendations, Volatility, a Stop/Loss a...
14 BBands with MM Rules	Long Only Upper Bollinger Band Breakout Entry with Money Management Exit Rules. Cl...
19 Boll Band & Average Cross E...	This strategy enters a long trade when the close crosses above the lower Bollinger Ban...
19 Boll Band & Call & Put Volum...	The Boll Band Reversal with Profit Target strategy builds on the basic Bollinger Band ...
19 Boll Band & MA Confirmation...	The Boll Band w MA Confirmation and Exit uses the middle average band of the Bollinge...
19 Boll Band & MACD Confirmati...	This strategy combines MACD as a filter on a basic Bollinger Band based reversal strate...
19 Boll Band & Max Loss Curren...	The Boll Band Reversal with Max Loss Currency strategy builds on the basic Bollinger ...
19 Boll Band & Max Loss Ticks	The Boll Band Reversal with Max Loss Ticks strategy builds on the basic CMCI by inclu...
19 Boll Band & Profit Target an...	The Boll Band Reversal with Profit Target and Stops strategy builds on the basic Bollin...
19 Boll Band & Profit Target Tic...	The Boll Band Reversal with Profit Target strategy builds on the basic Bollinger Band ...
19 Boll Band Reversal	All-in reversing Bollinger Band Strategy goes Long when the close crosses below the L...

Disclaimer: The strategies and data on BT <GO> and BTST <GO> are for illustrative purposes only, to demonstrate their functionality and possible data inputs. Users are responsible for selecting the particular parameters of any strategy they wish to monitor. Nothing on BT <GO> or BTST <GO> is meant to express an opinion on the future or projected value of any security, or to provide investment recommendations (i.e., as to whether or not to "buy", "sell", "hold", or to enter or not to enter into any transaction or employ any trading strategy). No aspect of BT <GO> or BTST <GO> is based on the consideration of a user's individual circumstances, and information available via BT <GO> and BTST <GO> should not be considered as information sufficient upon which to base an investment decision.

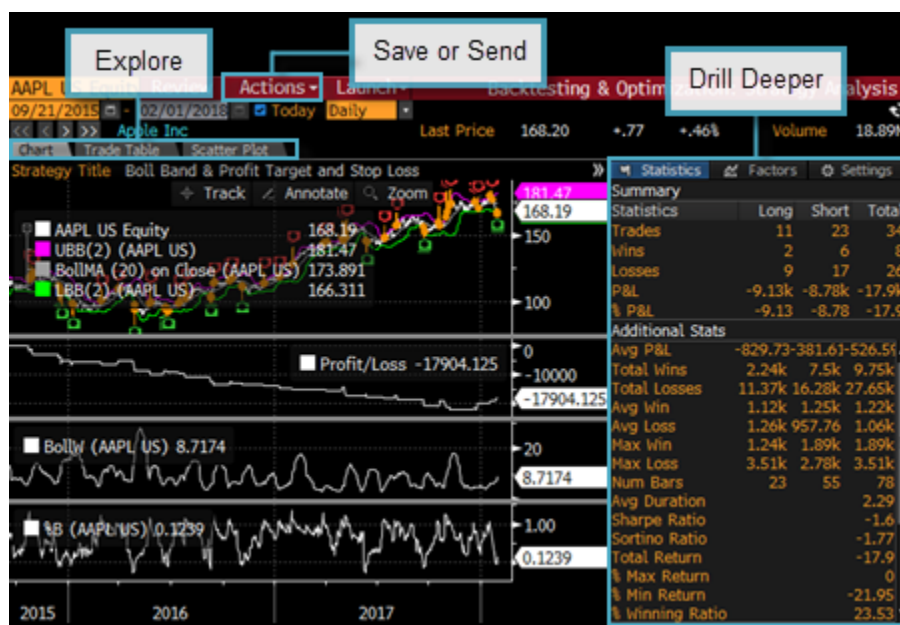
Quick Start

1. From the main screen, run a backtest with an existing strategy.



For more on creating your own strategy: [New Strategy](#).

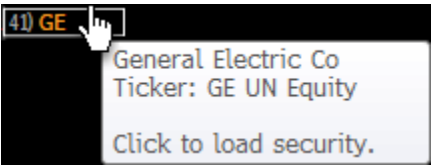
2. Start exploring and sharing the results of your backtest on the *Strategy Analysis* screen. Use the following map to guide you:



- **Explore:** Visualize the results of your analysis on the *Chart*, *Trade Table*, and *Scatter Plot* tabs. For more: [Single-Strategy Analysis](#).
- **Save or Send:** Save your results, export to Microsoft® Excel, or send your analysis via MSG. For more: [Sending Results](#).
- **Drill Deeper:** Get additional statistics or customize the way your results appear. For more: [Customizing Display Properties](#).

Getting Hints as You Go

On any Bloomberg function, you can mouse over key items for more information:



Workflows

Backtesting Basics

Backtesting allows you to assess the effectiveness of a trading strategy. Using BT <GO>, you can test a single strategy or multiple strategies across all asset classes for any number of securities. You can run backtests from the *Strategies*, *Multi-Strategy Worksheets*, or *Multi-Security Analyses* tabs on the main library screen.

Fast Backtesting

You can use the *Basic Backtesting* (BTST) function to run a pre-configured multi-strategy analysis for the currently-loaded security. If you're new to BT, this analysis is a good place to start.

To run BTST, enter BTST <GO>. The results of the test for the currently-loaded security appear in BT.

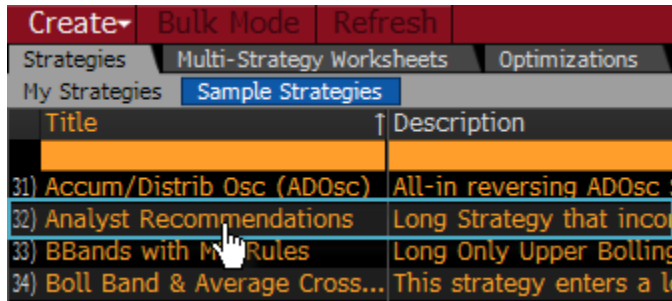
Worksheet Title BTST														
Strategy	Current Position			Trades			Profit (Loss)				Statistics			
	P&L	Age		Long	Short	Total	Total	%Total	Long	Short	Avg Duration	Profit Factor	SharpeSort	Rat
Maximum	2.71M	252		68	67	135	6.90M	6.90	2.71M	4.62M	252.00	1.97	1.55	1.65
Average	-25.27k	29		14	15	29.04	-8.26k	-0.01	-914.49	906.24	32.35	0.88	-0.05	-0.0
Median	0.00	7		11	11	22	0.00	0.00	-1.15M	509.35	11.57	0.84	-0.03	-0.0
Minimum	-2.45M	0		0	0	0	-3.34M	-3.34	-2.70M	-674.71	0.00	0.00	-0.82	-0.8
1) Buy & Hold	-2.45M	252		1	0	1	-2.45M	-2.45	-2.45M	0.00	252.00	0.00	-0.64	-0.6
2) Bollinger Bands (1											60.25	0.84	-0.23	-0.2
3) Cmdty Channel Inc											15.88	1.37	0.35	0.35
4) DMI											11.30	1.97	1.06	1.16
5) MACD											11.83	1.12	0.20	0.25
6) RSI													0.59	0.65
7) Stochastics (TAS)													-0.12	-0.1
8) William's %R (Wm													-0.50	-0.5
9) Parabolic (PTPS)													-0.64	-0.7
40) Simple MA (SMAvg													-0.61	-0.6
41) Exponential MA (E													0.23	0.25
42) Weighted MA (WMA													-0.65	-0.6
43) Variable MA (VMA													-0.69	-0.7
44) Triangular MA (TH													0.14	0.14
45) Accum (Distrib. Ov													-0.01	-0.0

[Hint] You can also run the BTST worksheet from the *Sample Worksheets* sub-tab of the main library's *Multi-Strategy Worksheets* tab. For more on the worksheet library: [Multi-Strategy Worksheet Library](#).

Running a Backtest

You can run a backtest from any of the libraries on the main library screen.

From the *Strategies*, *Multi-Strategy Worksheets*, or *Multi-Security Analysis* tab, select a library item (e.g., *Analyst Recommendations*).



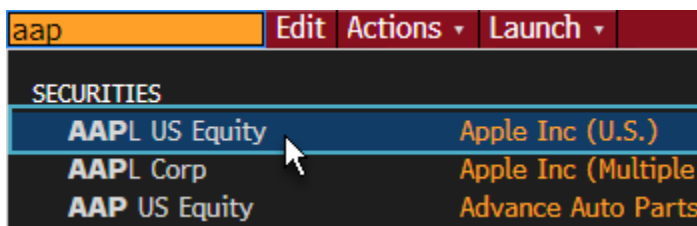
A "Running Simulation" message appears. Once the simulation finishes, the results of the backtest appear in either a chart or a table, depending on the type of test.

For more: [Single-Strategy Analysis](#), [Multi-Strategy Analysis](#), [Multi-Security Analysis](#).

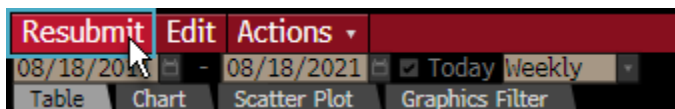
Updating the Security

You can change the loaded security(ies) directly from the results screen after you run a backtest, so you can continue working without navigating back to the main library screen.

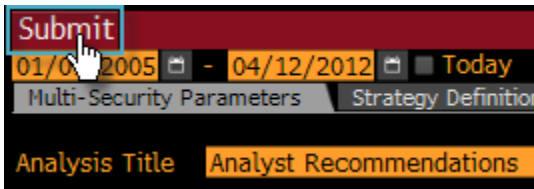
- To update the security from the *Strategy Analysis* or *Multi-Strategy* screen, in the amber search field, enter the name of a new security (e.g., **AAPL US Equity**) and select it from the list that appears.



- To update the security from the *Multi-Security Analysis* screen, from the toolbar, click **Resubmit**.



The *Resubmit Multi-Security Analysis* screen appears. Adjust your analysis, then click **Submit**.

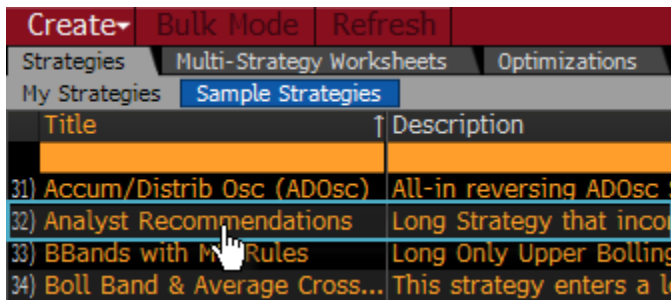


The backtest runs with the new security(ies).

Adjusting Date Range

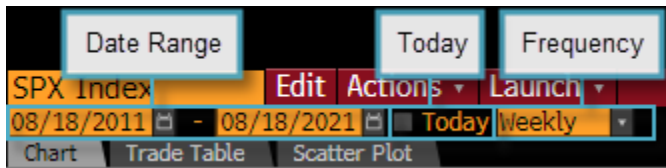
You can adjust the date range and frequency of the data used for a backtest.

1. Click any strategy (e.g., *Analyst Recommendations*).



The *Strategy Analysis* screen loads.

2. Update the date fields:



- **Date Range:** Edit the time window for the data you want to use in your backtest.
- **Today:** Select to indicate that you want to backtest real-time data, beginning with the date and time that you initiate the test. The data for the backtest is updated every 5 seconds.
- **Frequency:** Specify the frequency of the data (e.g., *Weekly*).

3. Press <GO>.

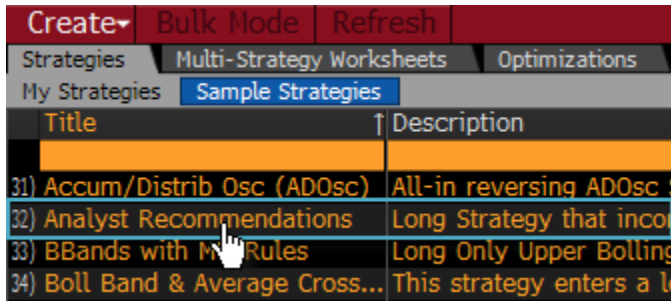
The date range updates.

Simulation Control

You can set simulation controls for single- or multi-strategy tests. Simulation controls are applied when you run the backtest and allow you to specify strategy parameters, such as trading approach, commission structure, and date range.

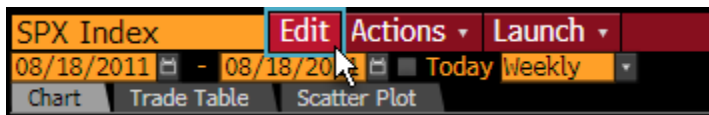
To set simulation controls for an existing strategy:

1. From the main library screen, select the strategy you want to adjust.



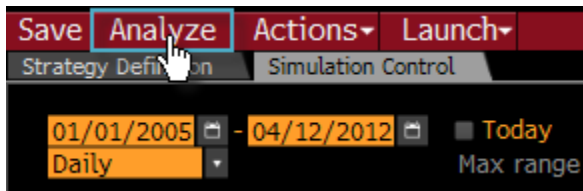
The *Strategy Analysis* screen appears.

2. From the toolbar, click **Edit**.



The *Strategy Definition* screen loads.

3. Select the *Simulation Control* tab, adjust any controls, then click **Analyze**.



Note: You can save your controls without running the simulation by clicking **Save**.

The *Strategy Analysis* screen appears with your new controls applied.

For more: [Simulation Controls](#).

Single-Strategy Analysis

You can create a new strategy to use in a backtest, or choose one from Bloomberg's library of samples. After you choose or define a strategy, you can visualize your results in a line chart, table, or scatter plot, so you can spot trends in your backtest.

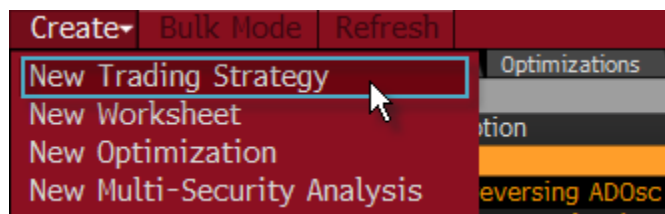
New Strategy

Defining a Strategy

You can create a new trading strategy with factors and rules that you define, so you can run a backtest using historical or intraday data.

Steps:

1. From the toolbar, click **Create > New Trading Strategy**.



The *Strategy Definition* screen appears.

2. In the *Strategy Title* field, enter the name you want to assign to the new strategy.
3. In the *Select From Available Factors* section, select one or more factors to include in your strategy (e.g., *Simple Moving Average*).



[Hint] To expand or collapse a category of factors, click the corresponding + icon. To expand or collapse the entire list, next to *Factors*, click the chevron.

The factor(s) appear in the *Selected Factors* section.

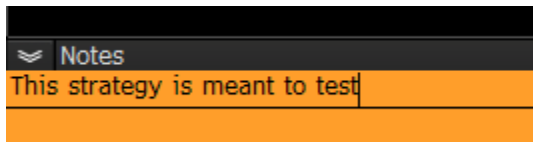
For more: [Adding Factors](#).

4. In the *Rules* section, define the entry/exit rules you want the strategy to test by making selections in the *Action*, *Factor*, and *Condition* fields.

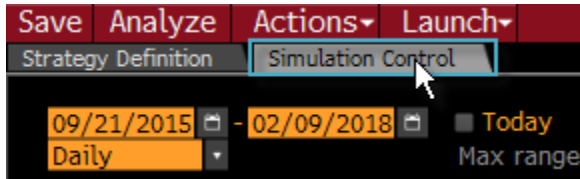


For more: [Defining Rules](#).

5. If you want to add notes to your strategy, next to *Notes*, click the chevron. Enter text in the amber field.

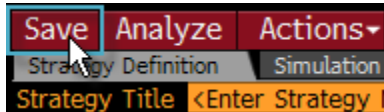


6. On the *Simulation Control* tab, adjust any controls you want to change.

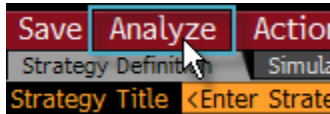


For more: [Simulation Controls](#).

7. To save your strategy, from the toolbar, click the **Save** button.



8. To run the simulation and see its results, from the toolbar, click **Analyze**.



The *Strategy Analysis* screen appears.

For more on analyzing results: [Results Analysis](#).

Adding Factors

You can add any combination of test factors to a strategy. Factors are metrics or conditions that BT tests against historical or intraday data, such as technical studies, individual securities, data fields, or events.

To add factors to your strategy:

1. Navigate to the *Strategy Definition* screen using the steps in [Defining a Strategy](#) or [Updating a Strategy](#).

The *Strategy Definition* screen appears.

2. In the *Available Factors* section, select a factor you want to include in your strategy (e.g., *Simple Moving Average*).

[Hint] To expand or collapse a category of factors, click the corresponding + button.

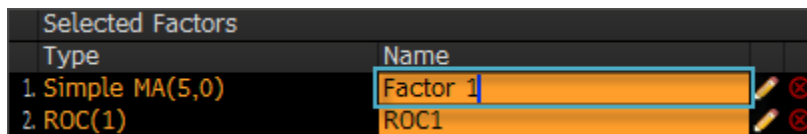


The factor appears in the *Selected Factors* section.

For more on adding events to a strategy: [Adding Events](#).

3. In the *Selected Factors* section, edit your factor(s):

- To name a factor, in its *Name* field, enter a name, then press <GO>.



- To edit the properties of a factor, including display properties, click the corresponding pencil icon.



The *Study Property Manager* window appears.

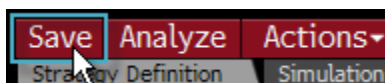
For more on editing display properties: [Customizing Display Properties](#).

- To delete a factor from the strategy, in the *Selected Factors* section, click the corresponding red X icon.



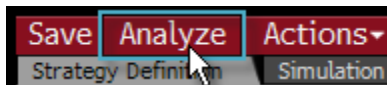
Your factor list updates with your selections. For more on individual studies: [TECH <GO>](#).

4. To save your strategy, from the toolbar, click the **Save** button.



A "Strategy successfully saved." message appears above the toolbar.

5. To run your simulation immediately, from the toolbar, click the **Analyze** button.



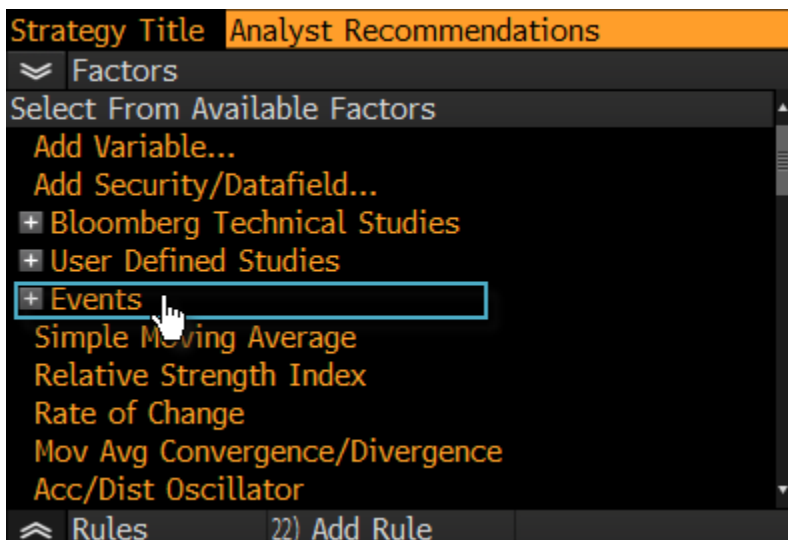
Your strategy results appear in the *Strategy Analysis* screen.

Adding Events

You can add real-world events, such as corporate actions and natural disasters, as factors in a trading strategy.

Steps:

1. From the *Strategy Definition* screen, in the *Select From Available Factors* section, select *Events*.



The *Events* tree appears.

2. Expand one of the sub-menus (e.g. *Corp Actions*), then select an event.

The event appears as a factor in the *Selected Factors* section.

Defining Rules

You must define at least one rule for a strategy in order for the strategy to be viable. Rules consist of trades that you want the backtest to simulate, market conditions that you want to trigger the trade, and the order type you want the test to execute when the condition is met.

For more on adding factors: [Adding Factors](#).

To define a rule:

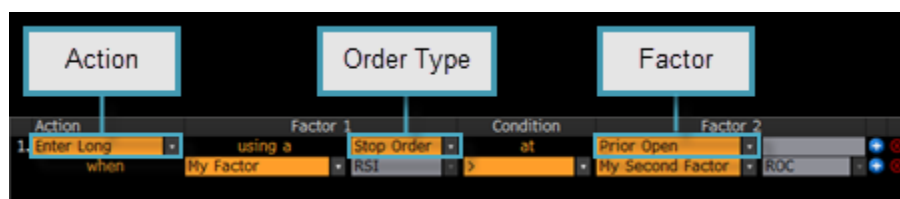
1. On the *Strategy Definition* screen, in the *Rules* section, click the **Add Rule** button.



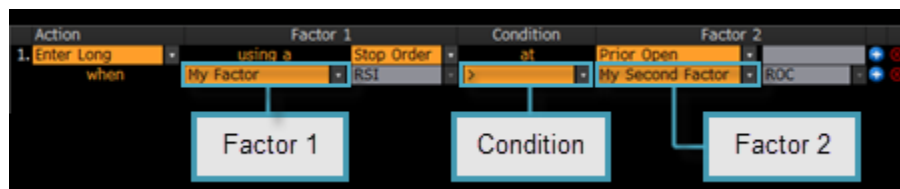
[Hint] To expand or collapse the *Rules* section, click the chevron next to *Rules*.

A new rule appears at the bottom of the rules list.

2. Define the action you want the strategy to simulate by selecting an action, order type, and factor in the rule's first row, as read from left to right.



- **Action:** Choose an action (e.g., *Enter Long*) that your backtest will simulate when the conditions you define are met.
 - **Order Type:** Specify the type of order (e.g., *Limit Order*) you want the trade to simulate .
 - **Factor:** Choose the action price (e.g., *Next Open*) for the entry or exit.
3. Define the trigger for the action you defined in the first row by selecting a first factor, condition, and second factor in the rule's second row, as read from left to right.



- **Factor 1:** Choose the first factor for comparison.
- **Condition:** Specify the relationship between Factor 1 and Factor 2 that triggers the action.
- **Factor 2:** Choose the second factor for comparison.

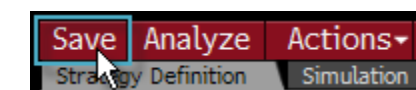
Note: Additional options may become available depending on the factors you select.

4. If you want to add more trigger criteria, click the + icon next to an action or trigger.



A new trigger row appears at the bottom of the rule's criteria list.

5. To save your rule, from the red toolbar, click **Save**.

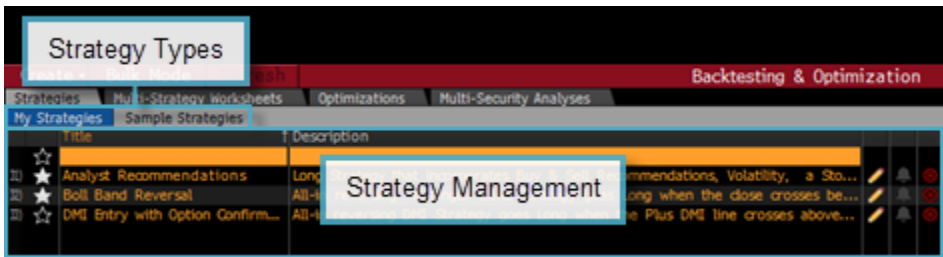


A "Strategy successfully saved." message appears above the toolbar.

Existing Strategies

Strategy Library

In the *Strategies* tab of the main library screen, your strategies are organized into two categories, so you can easily access custom and sample strategies.



- **Strategy Types:** Navigate to your strategies using two sub-tabs.
 - *My Strategies* contains all strategies you have created.
 - *Sample Strategies* contains strategies created and maintained by Bloomberg. You can use one of these samples as a starting point for your analysis.
- **Strategy Management:** Run, edit, or delete strategies, or create a strategy alert for one of your custom strategies.

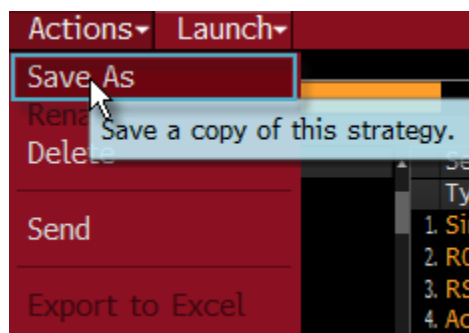
Note: Strategies you have marked as favorites appear with a highlighted star icon in the first column of the table.

Copying a Strategy

You can copy a strategy and use it as the basis for a new one.

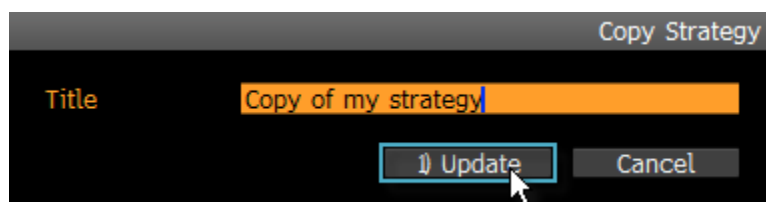
Steps:

1. On the *Strategy Definition* screen, from the toolbar, select **Actions > Save As**.



The *Copy Strategy* window appears.

2. In the *Title* field, enter a title for the copy, then click the **Update** button.



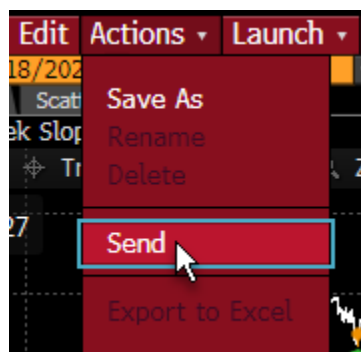
The strategy appears on the *My Strategies* sub-tab in *Strategy* tab of the main library screen.

Sending Results

Once you run a strategy, you can share it with other Bloomberg Professional® service users by sending it as a MSG <GO> attachment.

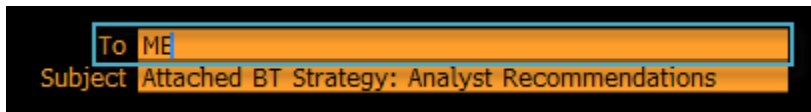
Steps:

1. On the *Strategy Analysis* screen, from the toolbar, select **Actions > Send**.



The *Send Strategy* window appears.

2. In the *To* field, enter the name of the recipient(s) and press <GO>.



[Hint] You can also edit the standard subject and message that appear.

3. If you want to exclude custom studies from the results you send, select *Hide Custom Studies*.



4. Click the **Send** button.



The strategy is sent to the recipient(s) as a MSG attachment.

Updating a Strategy

You can update an existing strategy's title, factors, rules, and simulation controls.

In the *Strategies* tab of the main library screen, click the pencil icon next to the strategy you want to edit.



The *Strategy Definition* screen appears.

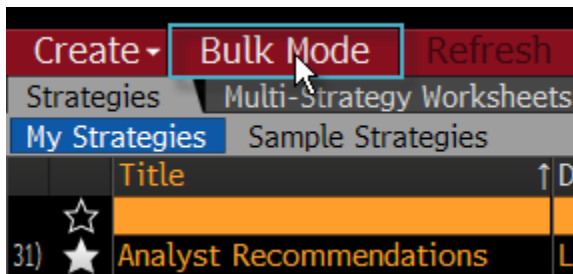
For more: [Defining a Strategy](#).

Bulk Mode

You can delete multiple strategies, worksheets, optimizations, or analyses at once using Bulk Mode.

Steps:

1. On the main library screen, from the toolbar, click the **Bulk Mode** button.



A delete column appears on the left side of the screen.

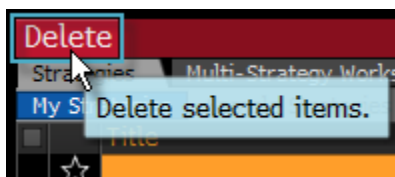
2. Navigate to the library tab that contains the items you want to delete (e.g., *My Strategies*).
3. In the delete column, select one or more library items to delete.



4. If you want to select all library items, select the box at the top of the column.



5. From the toolbar, click the **Delete** button.



6. In the *Warning* window that appears, click **Yes**.

Your strategies are deleted.

7. To exit Bulk Mode, press <Menu>.

Results Analysis

Single Strategy Chart

The *Chart* tab of the *Strategy Analysis* screen displays the trade events generated by the strategy simulation in a line chart, as well as key statistics and other study charts.

Note: A line chart appears for securities that do not have open, high, and low data.



- **Price Chart:** Visualize the security's price over the course of the backtest period. Each trade event triggered by your strategy is marked on the chart with green entry and exit icons.

Note: For a reversal strategy, the exit from a long trade is the entry for the next short trade.

- **Profit & Loss Chart:** See the total profit and loss incurred by the strategy you backtested over time.

Note: For a Buy and Hold strategy chart, the buy and hold P/L values depend on the period option you choose, since the trade dates are aligned with the beginning of the first period and the end of the last period. For example, if the dates are set to 2/5/09 2/5/10 and the period is set to Daily, the actual dates are 2/7/08 2/5/10; if Weekly, 2/11/08 2/2/10; if Monthly, 2/28/08 1/31/10.

- **Study Chart(s):** Visualize the results of individual factors in the backtest. You can adjust the factors that appear and their display properties in the *Factors* side panel.
- **Side Panel Navigation:** Navigate between the main *Statistics*, *Factors*, and *Settings* side panels.

- **Side Panel:** Get key statistics from your backtest. In the *Factors* side panel, choose factors to display and adjust display settings. In the *Settings* side panel, choose a security list to cycle through.

For more: [Single Strategy Table](#), [Single Strategy Scatter Plot](#).

Single Strategy Table

The *Trade Table* tab of the *Strategy Analysis* screen displays backtest results in a table format. The table includes information about entry and exit date/time and price as well as cumulative and per-trade statistics.

Side Panel Navigation											
Chart Trade Table Scatter Plot											
Strategy Title	Analyst	Recommendations									
Rule #	Action	Date	Price	Shares	Trade P&L	Cum. P&L	Total Return	% Max Return	% Min Return	% Max	Max Length
7	M/M	Sell	04/12/12	88.967	1859	54.37k	65.40k	48.97	52.19	.00	2.93
7	1	Buy	01/05/12	59.719	1859						
6	3	Sell	01/04/12	58.571	1895	3.10k	11.03k	2.87	3.18	.00	.96
6	1	Buy	12/22/11	56.936	1895						
5	3	Sell	12/13/11	56.143	1922	-4.24k	7.93k	-3.78	3.38	-10.98	13.89
5	1	Buy	10/13/11	58.347	1922						
4	3	Sell	10/12/11	58.191				15.14	16.87	-.15	12.44
4	1	Buy	07/12/11	50.536							
3	3	Sell	07/08/11	50.477				1.12	2.22	.00	.00
3	1	Buy	07/05/11	49.919							
2	3	Sell	05/31/11	48.729	1977	-2.36k	-3.66k	-2.39	.00	-4.94	4.94
2	1	Buy	05/10/11	49.921	1977						
1	3	Sell	05/05/11	49.771	1982	-1.31k	-1.31k	-1.31	.00	-1.91	1.91
1	1	Buy	04/25/11	50.43	1982						

Backtest Results

Zoom — — — — 100%

- **Side Panel Navigation:** In the *Statistics* side panel, get key statistics. In the *Columns* side panel, choose the columns that appear in the table. In the *Settings* side panel, choose a security list to cycle through.
- **Backtest Results:** Review the results of your backtest in table format. You can sort trades by column to easily spot extreme values.

Single Strategy Scatter Plot

You can plot two key metrics from your backtest results against each other on the *Scatter Plot* tab of the *Strategy Analysis* screen.



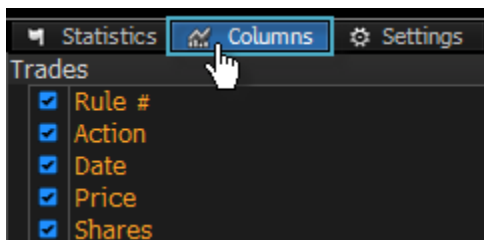
- **Side Panel Navigation:** Navigate between the *Statistics*, *Axes*, and *Settings* side panel.
- **Results Scatter Plot:** Compare two key metrics from your backtest. You can hover over any datapoint to see more information.
- **Side Panel:** Select the metrics you want to plot on the X and Y axes of the scatter plot. In the *Statistics* side panel, review key backtest statistics. In the *Settings* side panel, choose a security list to cycle through.

Customizing Display Properties

You can adjust the data that appears in the *Chart*, *Trade Table*, and *Scatter Plot* tabs from the corresponding side panel.

Steps:

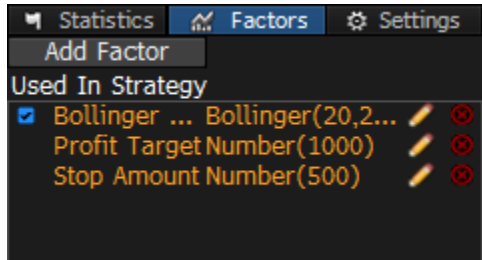
1. In the side panel navigation bar, select the middle option (e.g., on the *Trade Table* tab, click the **Columns** button).



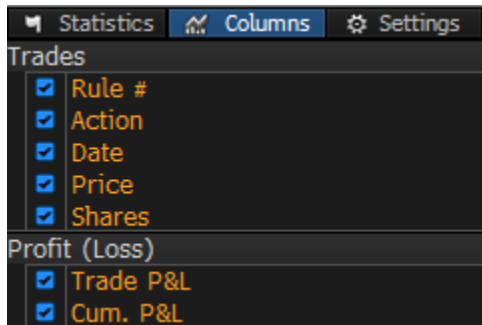
The tab's display options appear.

2. Customize the tab:

- On the *Chart* tab, edit parameters and display properties for the factors in the strategy. Click the **Add Factor** button to add a new factor to the chart.



- On the *Trade Table* tab, choose the columns you want to display.



- On the *Scatter Plot* tab, choose the metrics you want to compare on the X and Y axes.



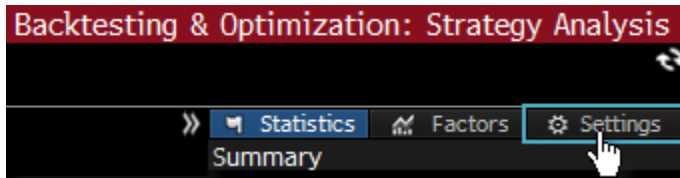
The chart, table, or scatter plot updates based on your selections.

Using a Security List

You can quickly cycle through backtest results for a custom list of securities after you run a single-strategy or multi-strategy analysis.

Steps:

1. After you run a single- or multi-strategy backtest, from the side panel navigation section, select the **Settings** button.



The *Settings* side panel appears.

2. From the *Source* drop-down menu, choose a list source (e.g., *Equity Index*).



The *Name* drop-down menu appears. Security list control buttons appear beneath the date fields.

Note: You can use one of your security worksheets as a security list. For more on security worksheets: [W Help Page](#).

3. Select a list of securities (e.g., *INDU*), then press <GO>.



The security list is loaded.

4. To cycle through the securities in your list, use the security list control buttons:



- To move to the first or the last security in the list, click the left- or right-facing chevron.
- To move to the next or previous security in the list, click the left- or right-facing single arrow.

Multi-Strategy Analysis

You can create a new multi-strategy worksheet to analyze multiple strategies at once, or choose one from Bloomberg's library of samples. Once you run a multi-strategy analysis from the *Multi-Strategy Worksheets* tab, you can analyze the results using a line chart, table, scatter plot, or graphics filter.

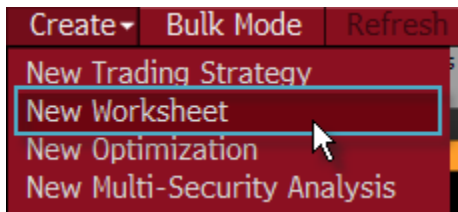
New Worksheets

Creating a Worksheet

You can create a worksheet that compares simulated trading results for a security across multiple strategies at once.

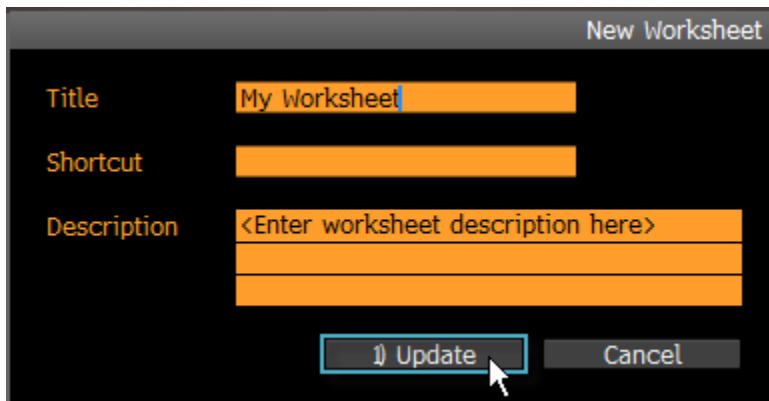
Steps:

1. In the main library screen, from the toolbar, select **Create > New Worksheet**.



The *New Worksheet* window appears.

2. Enter a title, shortcut, and description for the worksheet, then click **Update**.



A blank worksheet appears in the *Multi-Strategy* screen. By default, a Buy & Hold strategy appears as the first strategy on the worksheet.

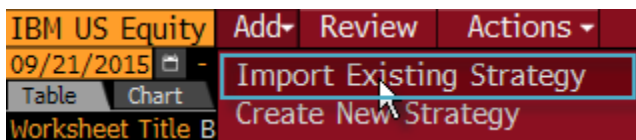
For more: [Multi-Strategy Chart](#), [Multi-Strategy Table](#), [Multi-Strategy Scatter Plot](#).

Adding a Strategy to a Worksheet

You can add any of your existing strategies to a worksheet, or create a new strategy right from the *Multi-Strategy* screen.

Steps:

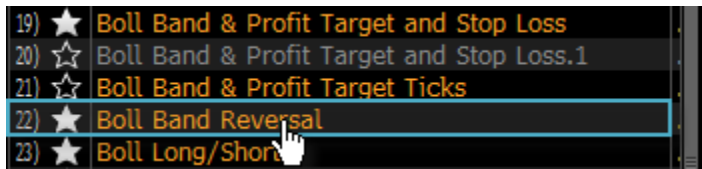
1. To add an existing strategy, in the *Multi Analysis* screen, select **Add > Import Existing Strategy**.



[Hint] You can also add a strategy to a worksheet from the side panel. In the *Strategies* sub-tab, click **Import Existing**.

The *Select Strategies* window appears.

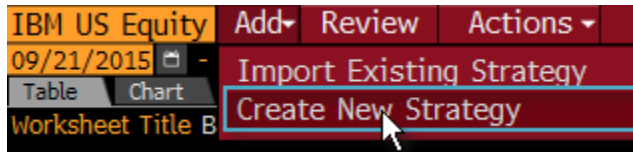
2. From the *My Strategies* or *Samples Strategies* tab, select one or more strategies (e.g., *Boll Band Reversal*), then click the **Update** button.



A "Running Simulation" message appears. Your worksheet appears with the new strategy(ies).

3. If you want to create a new strategy and add it to your worksheet:

a. From the red toolbar, select **Add > Create New Strategy**.



The *Strategy Definition* screen appears.

b. Follow the instructions in [Defining a Strategy](#) to create your strategy, then click the **Save** button.

A "Strategy successfully saved" message appears above the toolbar.

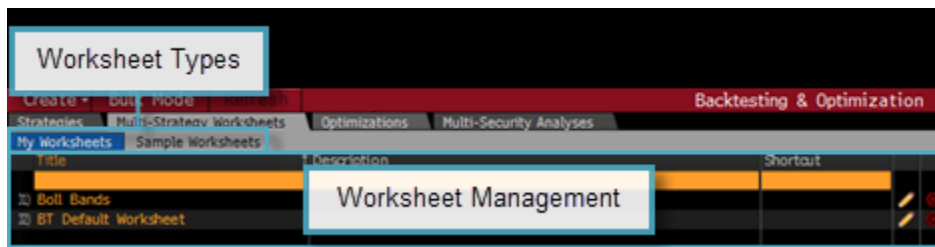
c. To return to your worksheet, press <Menu>.

Your worksheet appears with the new strategy added.

Existing Worksheets

Multi-Strategy Worksheet Library

In the *Multi-Strategy Worksheets* tab of the main library screen, your multi-strategy worksheets are organized into two categories, so you can easily access custom and sample worksheets.



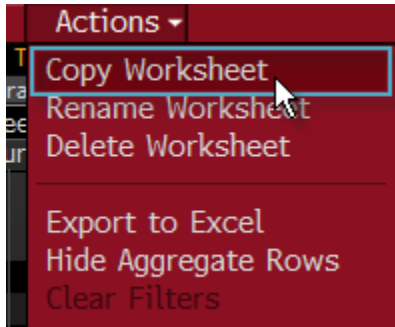
- **Worksheet Types:** Navigate to your worksheets using two sub-tabs.
 - *My Worksheets* contains all worksheets you have created.
 - *Sample Worksheets* contains worksheets created and maintained by Bloomberg. You can use one of these samples as a starting point for your analysis.
- **Worksheet Management:** Run, edit, or delete worksheets.

Copying a Worksheet

You can copy a worksheet and use it as the basis for a new one.

Steps:

1. On the *Multi-Strategy* screen, from the toolbar, select **Actions > Copy Worksheet**.



The *Copy Worksheet* window appears.

2. Enter a title, shortcut, and description for the new worksheet, then click the **Update** button.



The new worksheet appears.

Results Analysis

Multi-Strategy Table

By default, your multi-strategy analysis results appear in a table format, so you can easily compare strategies.

Side Panel Navigation

09/21/2015 - 09/22/2017 Today Daily

Table Chart Scatter Plot Graphics Filter

Worksheet Title Boll Bands

Strategies Columns Settings

Strategy		Current Position		Trades			Profit (Loss)		Statistics				
		P&L	Age	Long	Short	Total	Total	%Total	Long	Short	Avg Duration	Profit Factor	Sharpe Ratio
	Maximum	4.53k	505	32	30	62	9.33k	9.33	4.80k	4.53k	505.00	1.46	0.63 0.8
	Average	-2.66k	134	9	6	15.46	-807.35	-0.81	-1.97k	1.16k	130.51	0.66	0.04 0.0
	Median	-309.60	77	5	1	7	-587.33	-0.59	-629.9	1.16k	53.29	0.81	0.07 0.1
	Minimum	-14.13k	3	1	0	1	-15.93k	-15.93	-12.01	-8.39k	2.30	0.00	-0.57 -0.
21)	Golden Cross	4.53k	77	1	1	2	9.33k	9.33	4.80k	4.53k	174.50	0.00	0.49 0.6
32)	Golden Cross	4.53k	77	1	1	2	9.33k	9.33	4.80k	4.53k	174.50	0.00	0.49 0.6
33)	DMI with ADX Conf	1.16k	62.50	0.97	0.07	0.1							
34)	DMI with ADX Conf	1.16k	62.50	0.97	0.07	0.1							
35)	DMI Entry with Opt	-499.70	3	16	7	23	-4.11k	-4.11	831.64	3.28k	2.30	1.46	0.63 0.8
36)	CMCI & Profit Targ	-499.98	3	32	30	62	-7.58k	-7.58	-12.01	4.43k	2.39	0.81	-0.57 -0.
37)	CMCI & Profit Targ	-499.98	3	32	30	62	-7.58k	-7.58	-12.01	4.43k	2.39	0.81	-0.57 -0.
38)	Buy & Hold	442.24	505	1	0	1	442.24	0.44	442.24	0.00	505.00	0.00	0.10 0.1
39)	Boll Band Reversa	52.02	7	5	6	11	-15.93k	-15.93	-7.54k	-8.39k	45.64	0.64	-0.45 -0.
40)	Boll Band Reversa	442.24	505	1	0	1	442.24	0.44	442.24	0.00	505.00	0.00	0.10 0.1
41)	Analyst Recomm	-14.13k	183	7	0	7	-629.94	-0.63	-629.9	0.00	53.29	0.96	0.03 0.0
42)	Analyst Recomm	-14.13k	183	7	0	7	-629.94	-0.63	-629.9	0.00	53.29	0.96	0.03 0.0
43)	Analyst Recomm	-14.13k	183	7	0	7	-629.94	-0.63	-629.9	0.00	53.29	0.96	0.03 0.0

Multi-Strategy Analysis Results

Zoom 100%

- **Side Panel Navigation:** In the *Strategies* side panel, choose the strategies you want to include in the table. In the *Columns* side panel, choose the columns you want to include in the table. In the *Settings* side panel, choose a security list you want to cycle through.
- **Multi-Strategy Analysis Results:** Review the results of your backtest, sort data, set alerts, and manage strategies. For more on defining strategies: [Defining a Strategy](#).

For more on setting alerts: [Strategy Alerts](#).

Multi-Strategy Chart

You can compare strategies in a visual format on the *Charts* tab of the *Multi-Strategy* screen.



- **Price Chart:** Visualize the price of the loaded security over the course of the backtest.
- **Individual Study Chart:** Compare the performance of individual strategies. You can choose the strategies that appear from the *Strategies* side panel, or adjust the metric from the *Metrics* side panel.
- **Cumulative Study Chart:** Visualize the cumulative performance of all the strategies selected in the *Strategies* side panel, or adjust the metric using the *Metrics* side panel.
- **Side Panel Navigation:** Navigate between the *Strategies*, *Metrics*, and *Settings* tabs.
- **Side Panel:** Select strategies to include in the chart. In the *Metrics* tab, toggle the metric displayed in the charts. In the *Settings* tab, choose a security list you want to cycle through.

Multi-Strategy Scatter Plot

You can compare strategies using two metrics that you specify on the *Scatter Plot* tab of the *Multi-Strategy* screen.



- **Scatter Plot:** Visualize your backtest results. You can adjust the view using the **Zoom** and **Track** tools or copy the chart using the **Copy** button.
- **Side Panel Navigation:** Navigate between the *Strategies*, *Axes*, and *Settings* side panels.
- **Side Panel:** Choose the metrics you want to compare on the X and Y axes in the scatter plot. In the *Strategies* side panel, select strategies to include in the scatter plot. In the *Settings* side panel, choose a security playlist to cycle through.

Multi-Strategy Graphics Filtering

You can use graphics filtering to visualize combinations within your backtest results, so you can easily compare metrics across securities or strategies.

For more: [Graphics Filtering](#).

Multi-Security Analysis

You can test a strategy against multiple securities at once. Once you run a multi-security analysis from the *Multi-Security Analyses* tab, you can visualize the results in a line chart, table, scatter plot, or graphics filter.

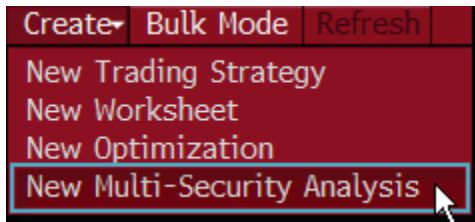
New Analyses

Creating a Multi-Security Analysis

You can create a new multi-security analysis to analyze the effectiveness of a strategy across multiple securities.

Steps:

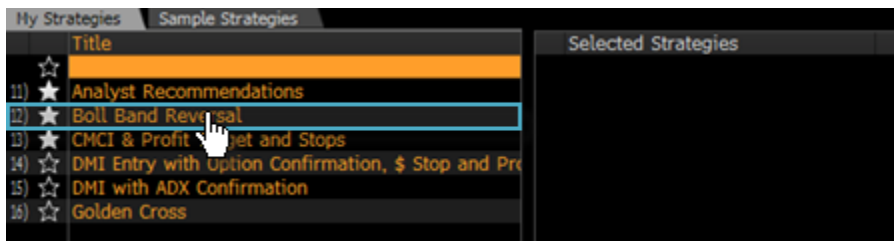
1. From the toolbar, select **Create > New Multi-Security Analysis**.



[Hint] You can easily transform a single-strategy test for one security into a multi-security test from the *Strategy Analysis* screen. From the toolbar, select **Launch > Multi-Security Analysis**.

The *Select Strategy for Multi-Security Analysis* window appears.

2. From the *My Strategies* or *Sample Strategies* tab, select the strategy you want to backtest (e.g. *Boll Band Reversal*).



Note: You can test only one strategy at a time against a group of securities.

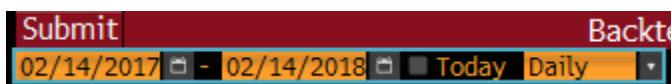
The strategy appears in the *Selected Strategies* section.

3. Click the **Update** button.



The *Submit Multi-Security Analysis* screen appears.

4. Choose a date and periodicity for the analysis.



[Hint] On the *Strategy Definition* tab, you can check the definition of the strategy you chose.

5. In the *Analysis Title* field, enter a name for the analysis.

Analysis Title

6. To choose a security list, in the *Analysis Source* section, select a list source and the name of the list you want to use.

Analysis Source

Source

Name

7. To adjust simulation controls, navigate to the *Simulation Control* tab.

For more: [Simulation Controls](#).

8. From the toolbar, click the **Submit** button.

Submit

02/14/2017 - 02/14/2018

A "Loading Multi-Security Analysis..." window appears. The test results appear in the *Multi-Security Analysis* screen and are sent to your inbox as a MSG attachment.

For more: [Results Analysis](#), [Updating a Multi-Security Analysis](#).

Existing Analyses

Multi-Security Analyses Library

In the *Multi-Security Analyses* tab of the main library screen, your multi-security analyses are organized into three categories, so you can easily access current, past, and sample analyses.

Analysis Types

Create | Multi-Strategy Worksheets | Optimizations | Multi-Security Analyses

Active Analyses | Expired Analyses | Sample Analyses

Analysis Title	Strategy Title	List Name	Start Date	End Date	Period	Status	Submitted
1) Boll Band Rev...	Boll Band Rever...				Daily	67% Done	13:50:24
2) Analyst Recom...	Analyst Recom...				Daily	Complete	09/22/2017

Analysis Management

- **Analysis Types:** Navigate to your worksheets using three sub-tabs.
 - *Active Analyses* contains all analyses created within the last thirty days. Analyses that have not finished running appear in green.
 - *Expired Analyses* contains analyses created more than thirty days ago.
 - *Sample Analyses* contains analyses created and maintained by Bloomberg. You can use one of these samples as a starting point for your analysis.
- **Analysis Management:** Access analysis results or delete analyses.

Updating a Multi-Security Analysis

You can update and resubmit a multi-security analysis you have already submitted.

Steps:

1. On the *Multi-Security Analysis* screen, from the toolbar, click the **Resubmit** button.



The *Resubmit Multi-Security Analysis* screen appears.

2. Follow the instructions in [Creating a Multi-Security Analysis](#) to update your analysis.
3. From the toolbar, click the **Submit** button.



The analysis is run with the updated parameters. The test results appear in the *Multi-Security Analysis* screen and are sent to your inbox as a MSG attachment.

Results Analysis

Multi-Security Table

By default, your multi-security analysis results appear in a table format, so you can assess the effectiveness of a strategy across multiple securities.

Column Selection

Resubmit	Review	Actions	Backtesting & Optimization													
10/10/2012	10/10/2013	Today	Daily													
Table	Chart	Scatter Plot	Graphics Filter													
Analysis Title			Boll Band Reversal													
			Columns													
			Current Position			Trades			Profit (Loss)			Statistics				
Security			P&L	Age	Long	Short	Total	Total	%Total	Long	Short	Avg Duration	Profit Factor	Sharpe Ratio	Sortino Ratio	Total Return
Maximum			360.50k	249	1	0	1	360.50k	360.50k	360.50k	0.00	250.00	0.00	13.40	39.02	360.50
Average			28.00k	245	1	0	1	28.00k	28.00k	28.00k	0.00	245.71	0.00	1.81	3.29	28.00
Median			23.97k	247	1	0	1	23.97k	23.97k	23.97k	0.00	248.00	0.00	1.66	2.80	23.97
Minimum			-52.24k	61	1	0	1	-52.24k	-52.24k	-52.24k	0.00	62.00	0.00	-1.34	-2.11	-52.24
1) ABT UN Equity			-2.69k	246	1	0	1	-2.69k	-2.69k	-2.69k	0.00	247.00	0.00	-0.07	-0.11	-2.69
2) A UN Equity			34.22k	245	1	0	1	34.22k	34.22k	34.22k	0.00	246.00	0.00	2.16	3.74	34.22
3) ACN UN Equity			2.11k	249	1	0	1	2.11k	2.11k	2.11k	0.00	250.00	0.00	0.25	0.36	2.11
4) AAP UN Equity			20.05k	246	1	0	1	20.05k	20.05k	20.05k	0.00	247.00	0.00	1.23	2.60	20.05
5) ADBE UN Equity			59.94k	245	1	0	1	59.94k	59.94k	59.94k	0.00	246.00	0.00	3.98	9.19	59.94
6) DGX UN Equity			-8.92k	246	1	0	1	-8.92k	-8.92k	-8.92k	0.00	247.00	0.00	-0.49	-0.67	-8.92
7) AAPL UN Equity			-24.24k	249	1	0	1	-24.24k	-24.24k	-24.24k	0.00	250.00	0.00	-0.72	-1.08	-24.24
8) ABBV UN Equity			29.58k	208	1	0	1	29.58k	29.58k	29.58k	0.00	209.00	0.00	2.01	3.34	29.58
9) DFS UN Equity			29.13k	246	1	0	1	29.13k	29.13k	29.13k	0.00	247.00	0.00	1.88	3.24	29.13
10) DG UN Equity			14.30k	245	1	0	1	14.30k	14.30k	14.30k	0.00	246.00	0.00	0.86	1.37	14.30
11) ABC UN Equity			56.08k	246	1	0	1	56.08k	56.08k	56.08k	0.00	247.00	0.00	4.81	8.71	56.08
12) JWN UN Equity			1.58k	246	1	0	1	1.58k	1.58k	1.58k	0.00	247.00	0.00	0.21	0.35	1.58
13) JPM UN Equity			24.03k	249	1	0	1	24.03k	24.04k	24.03k	0.00	250.00	0.00	1.69	2.77	24.04
14) IVZ UN Equity			34.87k	245	1	0	1	34.87k	34.87k	34.87k	0.00	246.00	0.00	2.10	3.69	34.87
15) BIL UN Equity			6.77k	246	1	0	1	6.77k	6.77k	6.77k	0.00	247.00	0.00	0.61	0.98	6.77

- **Column Selection:** Choose the columns that appear in your results.
- **Multi-Security Analysis Results:** See the results of your backtest. Click a security to examine its results in the *Strategy Analysis* screen.

Multi-Security Chart

You can visualize the results of a strategy across multiple securities on the *Charts* tab of the *Multi-Security Analysis* screen.



- **Results Chart:** Visualize aggregate Profit/Loss or Trade Size over the course of the backtest.
- **Side Panel Navigation:** Navigate between the *Securities* and *Metrics* side panels.
- **Side Panel:** In the *Securities* tab, select the securities included in the results chart. In the *Metrics* tab, toggle the metric that appears in the results chart.

Multi-Security Scatter Plot

You can plot securities against two metrics that you specify on the *Scatter Plot* tab of the *Multi-Security Analysis* screen .



- **Side Panel Navigation:** Navigate between the *Securities* and *Axes* tabs.
- **Scatter Plot:** Compare securities using two metrics that you specify in the side panel. You can adjust the view using the *Zoom* and *Track* tools or copy the chart using the *Copy* button.
- **Side Panel:** Choose the metrics used in the scatter plot. In the *Securities* tab, select securities to include in the scatter plot.

Editing Analysis Parameters

You can update the underlying security list, strategy, and simulation controls of a multi-security analysis.

On the *Multi-Security Analysis* screen, from the toolbar, click the **Edit** button.



The *Analysis Definition* screen appears.

Exporting an Analysis

You can export a multi-security analysis to Microsoft® Excel.

On the *Multi-Security Analysis* screen, from the toolbar, select **Actions > Export**.



Your results appear in an Excel spreadsheet.

Optimizing Parameters

You can optimize the parameters used in a strategy you are backtesting, so you can determine the parameter values that achieve the best results for your test.

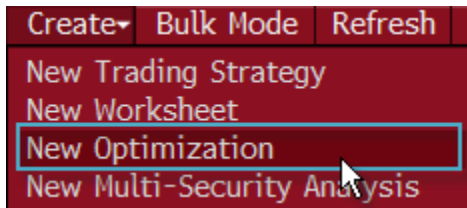
New Optimizations

Creating an Optimization

You can optimize any of the parameters used in a strategy.

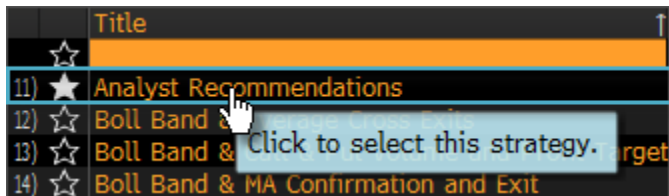
Steps:

1. On the main library screen, from the toolbar, select **Create > New Optimization**.



The *Select Strategy to Optimize* window appears.

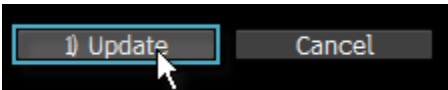
2. From the *My Strategies* or *Sample Strategies* tab, select the strategy for which you want to optimize parameters (e.g. *Analyst Recommendations*).



The strategy appears in the *Selected Strategies* section.

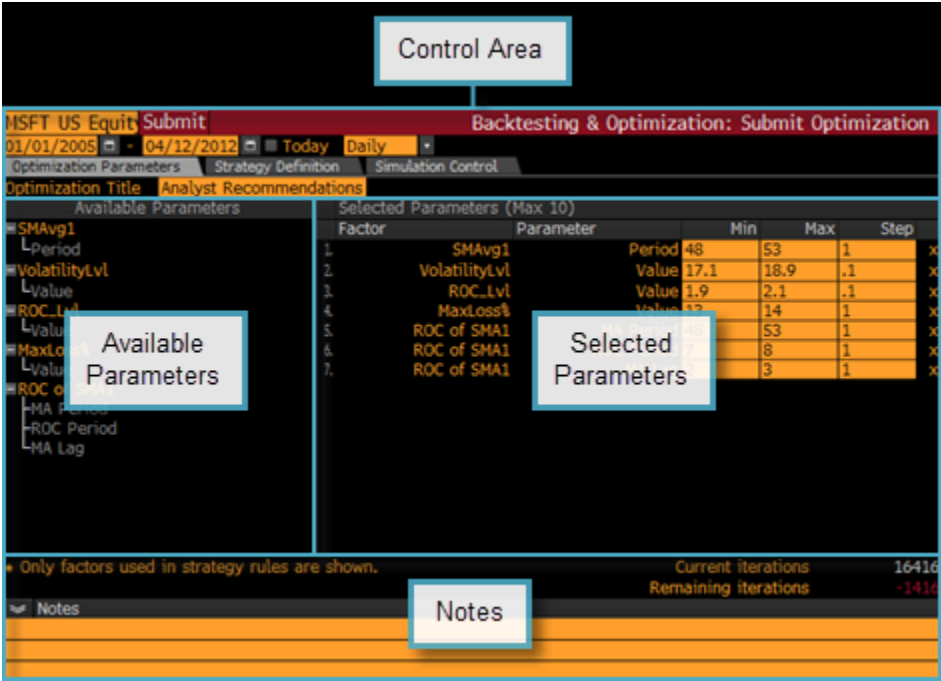
Note: You can only optimize one strategy at a time.

3. Click the **Update** button.



The *Submit Optimization* screen appears with default parameters defined for the selected strategy.

4. Define your optimization:

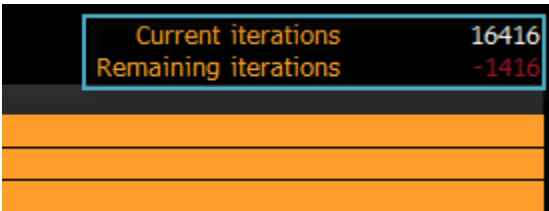


- **Control Area:** Choose a date, periodicity, and name for your optimization.
- **Available Parameters:** Select the strategy parameters you want to optimize.
- **Selected Parameters:** In the **Min**, **Max**, and **Step** fields, enter values for each parameter.
Note: To remove a parameter from your list, click the corresponding yellow **X** icon.
- **Notes:** Enter explanatory comments for the optimization.

[Hint] You can review the underlying strategy or adjust simulation controls from the *Strategy Definition* or *Simulation Control* tabs.

For more: [Simulation Controls](#).

5. Above the *Notes* section, check the *Current iterations* field. If the iteration count is above 15,000, in the *Selected Parameters* section, delete one or more parameters from your list.

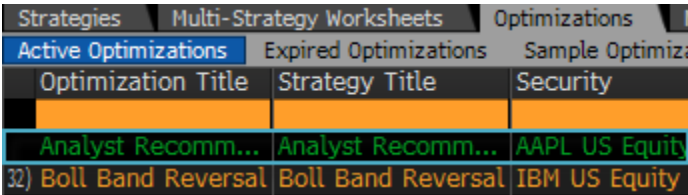


You *Current iterations* count updates.

6. From the toolbar, click the **Submit** button.



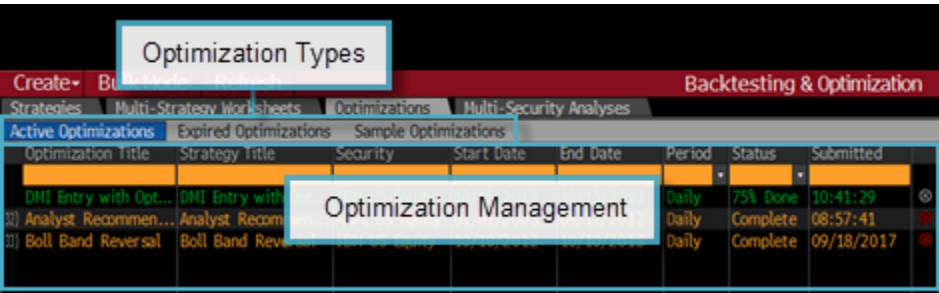
The optimization appears in the main library screen.



Existing Optimizations

Optimization Library

In the *Optimizations* tab of the main library screen, your optimizations are organized into three categories, so you can easily access current, past, and sample optimizations.



- **Optimization Types:** Navigate to your optimizations using three sub-tabs:
 - *Active Optimizations* contains all optimizations submitted within the last 30 days.
 - *Expired Optimizations* contains all optimizations that were submitted more than 30 days ago. Once an optimization is expired, you cannot drill down into the details for each optimization pass. You can still display the table and 3D surface charts of the analysis, as well as the *Strategy Definition* and *Simulation Control* tabs, where you can see the settings that were in effect when the optimization was last run.

- *Sample Optimizations* contains optimizations created and maintained by Bloomberg. You can use one of these samples as a starting point for your analysis.
- **Optimization Management:** Click an optimization to see results, or delete active, expired, and sample optimizations.

Note: Optimizations that have not finished running appear in green.

For more: [Optimization Results](#).

Optimization Results

Analyzing Optimization Results

Once you submit an optimization, you can see the results in a table, 3D chart, or graphics filter.

Steps:

1. From the *Optimizations* tab, select a completed optimization from the *Active Optimizations*, *Expired Optimizations*, or *Sample Optimizations* sub-tabs.

Backtesting & Optimization						
Create		Bulk Mode	Refresh			
Strategies		Multi-Strategy Worksheets		Optimizations	Multi-Security Analyses	
Active Optimizations		Expired Optimizations		Sample Optimizations		
Optimization Title	Strategy Title	Security	Start Date	End Date	Period	
Stochastics (TAS)	Stochastics (TAS)	SPY US Equity	01/03/2011	11/15/2016	Daily	
DMI	DMI	SIXR Index	01/01/2000	07/29/2016	Weekly	
MACD	MACD	SXSE Index	01/01/2000	07/28/2016	Weekly	

The results of the optimization appear. You can use the horizontal scroll bar at the bottom of the table to see all of the data columns.

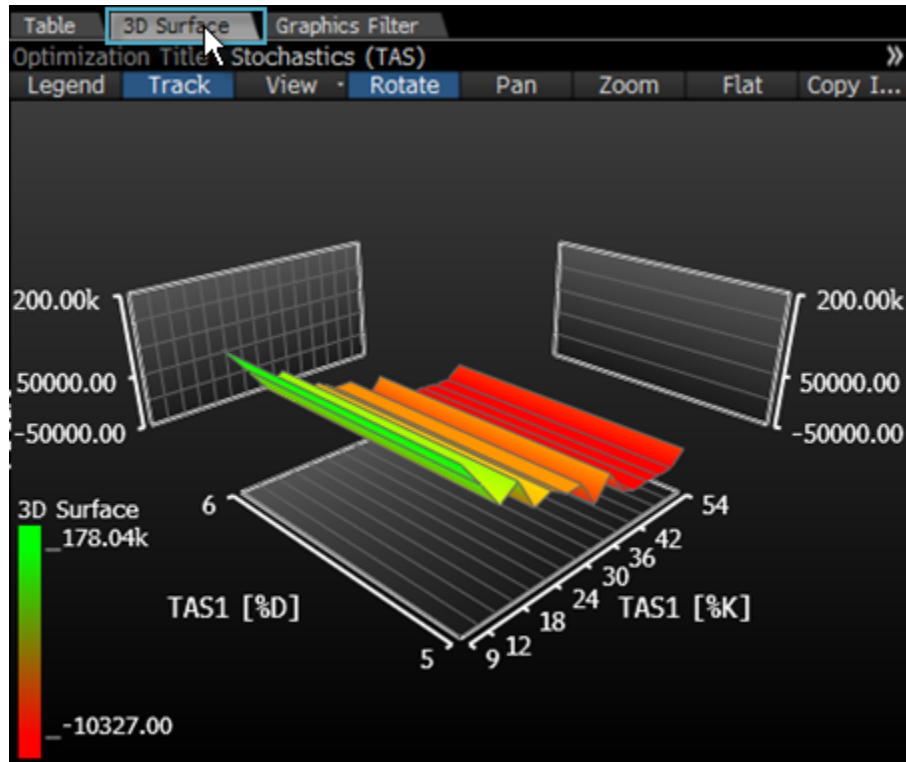
For more on the data columns: [Definitions](#).

2. Click any data column to sort the table and find the highest-performing parameter combination. For example, sort by *Total* to display the most profitable combination at the top.

Trades		Profit (Loss)		Statistics	
Long	Short	Total	%Total	Long	Short
93	93	186	193.14k	193.14	143.72
35	35	70.34	40.05k	40.05	51.95k
27	27	54	30.71k	30.71	46.35k
16	16	32	-29.45k	-29.45	8.71k
65	65	130	193.14k	193.14	143.72
65	65	130	193.14k	193.14	143.72
65	65	130	193.14k	193.14	143.72
65	65	130	193.14k	193.14	143.72
65	65	130	193.14k	193.14	143.72

3. Further your analysis:

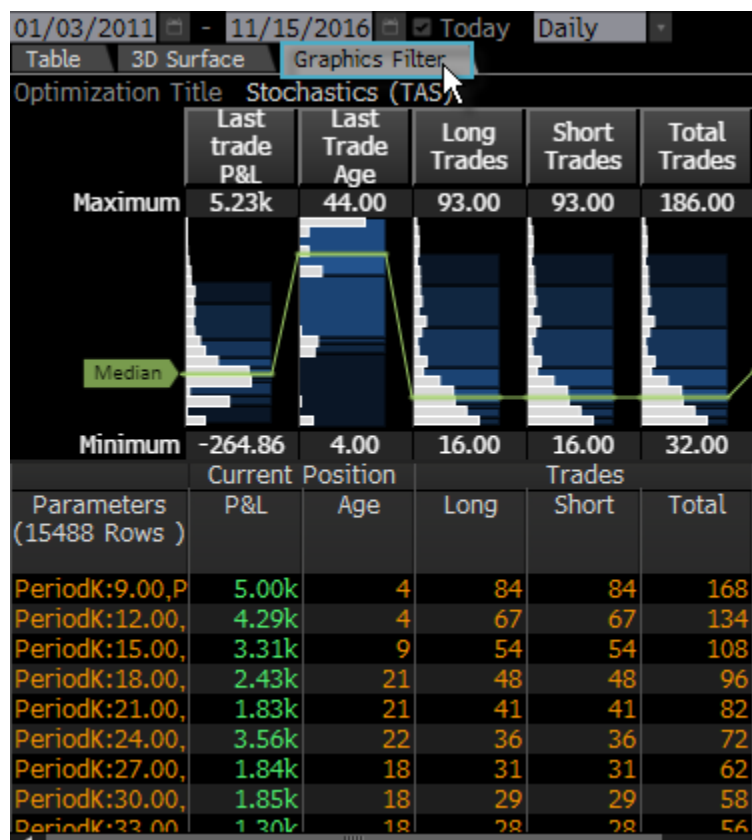
- If you want to analyze the results in a three-dimensional chart, click the **3D Surface** tab.



The data appears in a 3D chart.

For more: [Optimization 3D Chart](#).

- If you want to analyze the results using a graphical representations of each parameter's range, click the **Graphics Filter** tab.

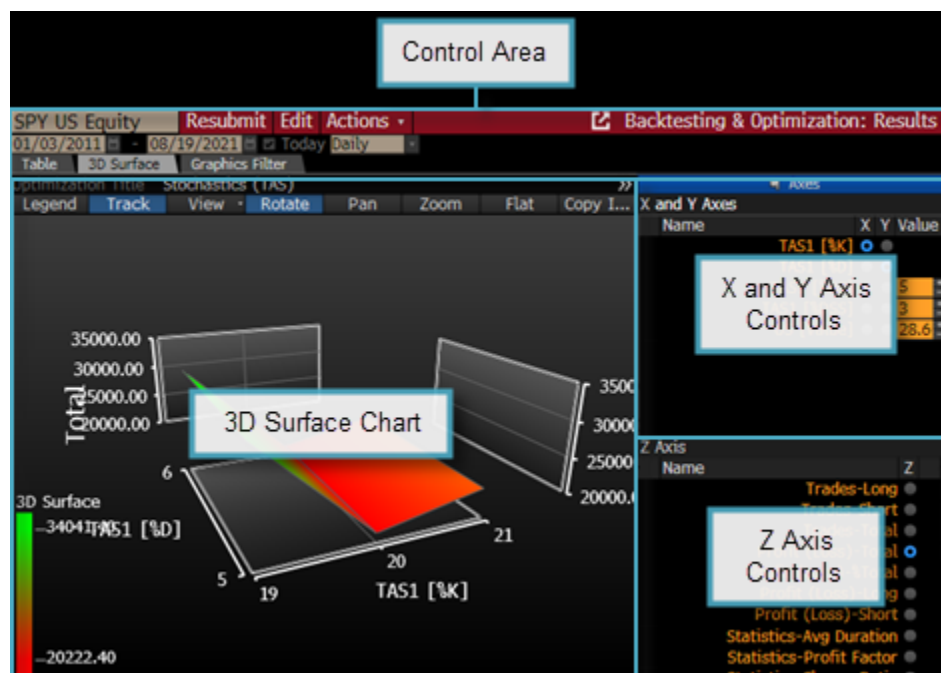


The data appears in a graphics filter.

For more: [Graphics Filtering](#).

Optimization 3D Chart

On the *Results* screen, you can use the *3D Chart* tab to visualize the relationship between two parameters in an optimization.



- **Control Area:** Resubmit the optimization, review the underlying parameters and strategy, or navigate to the *Table* or *Graphics Filter* tabs.
- **3D Surface Chart:** Visualize the optimization results in a 3D format. Change the perspective of the chart using the *View*, *Rotate*, *Pan*, or *Zoom* tools, or display the data in a contour plot using the *Flat* tool.
- **X and Y Axis Controls:** Choose two parameters to plot on the X and Y axes. Use the + and - buttons to adjust the values of other parameters.
- **Z Axis controls:** Choose the metric or statistic to plot on the Z axis.

Optimization Graphics Filtering

You can use graphics filtering to visualize combinations within your optimization results, so you can easily compare metrics.

For more: [Graphics Filtering](#).

Graphics Filtering

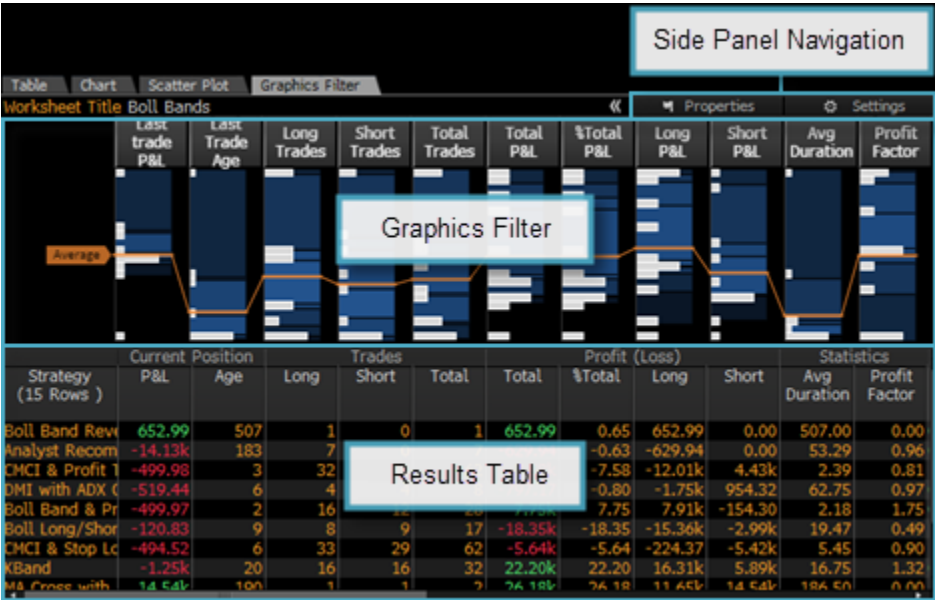
You can visualize comparisons of key metrics across securities or strategies in your backtest results. Graphics filtering is available on the results screens for multi-strategy backtests, multi-security backtests, and optimizations.

Graphics Filtering Basics

Graphics Filter Tab

You can use graphics filtering on the results screens for multi-strategy backtests, multi-security backtests, and optimizations.

The *Graphics Filter* tab is divided into a side panel navigation bar, a graphics filter, and a table of test results.



- **Side Panel Navigation:** In the *Properties* tab, select the data that appears in the graphics filter. In the *Settings* tab, select a security list to cycle through.
- **Graphics Filter:** Select a range for one or more metrics with which to filter the data table. You can adjust the data that appears in the graphics filter on the *Properties* tab.
- **Results Table:** Click a security or strategy to highlight its results in the graphics filter.

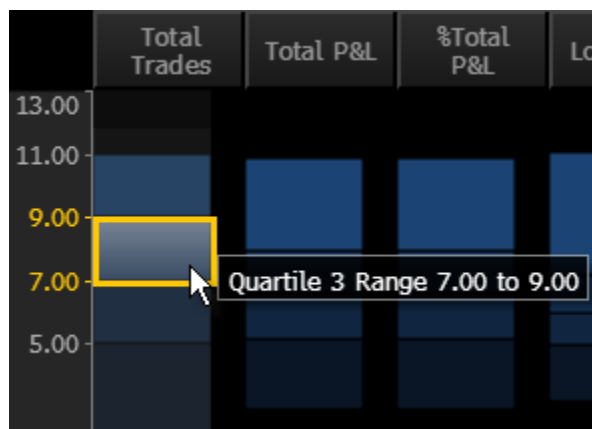
Filtering Test Results

You can isolate ranges of data inside or across quantiles , so you can create subsets of data to analyze and compare.

To filter the data table using a results column:

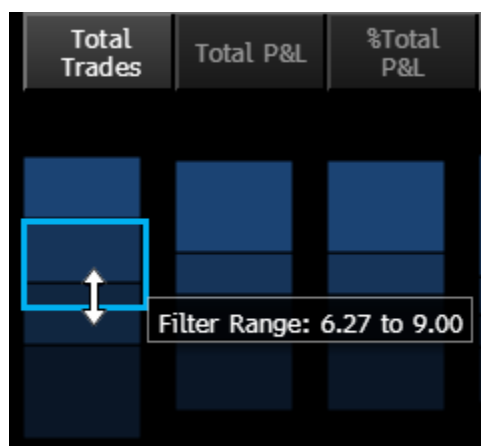
1. In the column for the metric you want to use to filter (i.e., *Total Trades*), position your mouse over a [quantile](#), then left-click.

[Hint] A yellow box appears around individual quantiles as you move your mouse up and down the data column.



A blue box appears around the **quantile** and the data in the table below is filtered.

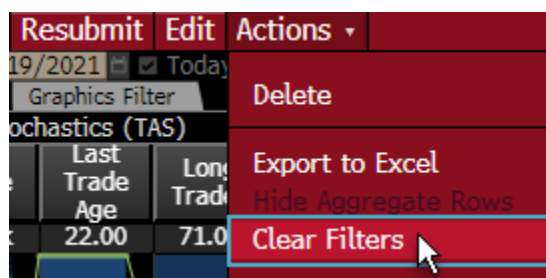
- If you want to refine the data range, position your mouse over the top or bottom edge of the blue box, then click and drag the box until the range that you want to focus on is surrounded.



[Hint] You can also click and drag the blue box to move it around the column.

The data in the table below is filtered based on the range you selected.

- To clear the filters you created, from the red toolbar, select **Actions > Clear Filters**.



All filters are removed.

Sorting Test Results

You can sort individual data columns in the results table in ascending or descending order.

In the table, click the heading of the column that you want to sort.

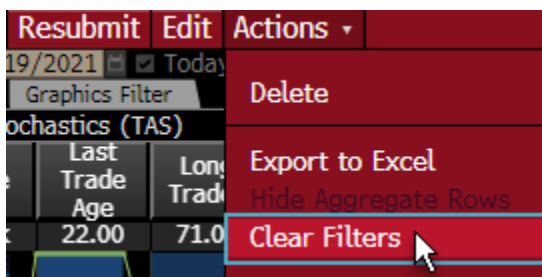
Security (20 Rows)	Current Position	
	P&L	Age
USSWAP10 CBBT	15.64k	19
USSWAP4 CBBT	14.84k	160
USSWAP3 CBBT	42.66k	25
USSWAP20 CBBT	9.61k	4
USSWAP2 CBBT	16.31k	25
USSP10 Curncy	10.82k	47

The data in the column appears in either ascending or descending order.

Clearing Filters

While filtering the test results, you can start over with a new analysis by clearing the filters.

To clear filters you have applied, from the toolbar, select **Actions > Clear Filters**.



All filters are removed.

Further Analysis

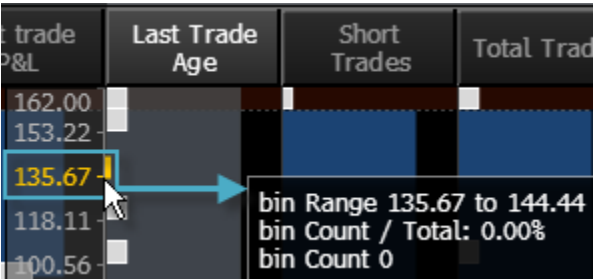
Displaying Additional Data

You can display information about how the data is filtered on the graph, as well as summary statistics across all columns.

Steps:

1. Position your mouse over the data for which you want more information:

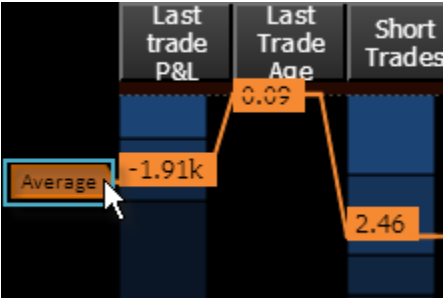
- To display the **bin** range for an individual data point, position your mouse over the left margin of a column. Then, move your mouse up and down the margin until the value you want information for appears in yellow.



- To highlight the median value across all columns, position your mouse over the *Median* label. The median value in all of the columns is highlighted.

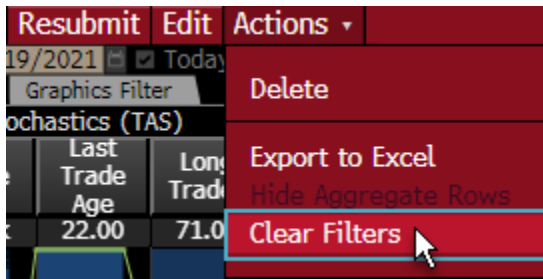


- To highlight the average value across all columns, position your mouse over the *Average* label. The average value in all of the columns is highlighted.



Note: You can control the appearance of the median and average tags by using the settings in the *Properties* side panel.

2. To clear the filters you created, from the red toolbar, select **Actions > Clear Filters**.



All filters are removed.

Setting Filter Properties

You can manage the way test results appear in the graphics filter from the *Properties* side panel. You can choose how the data is grouped, specify which summary statistics appear, or add a distribution histogram to the data.

Select	To
A quantile grouping method (e.g., <i>Quartile</i>)	Specify how you want to group the test result data. For more: Interpreting Quantiles .
Distributions	Add a distribution histogram to each column.
Gutters	Spread out the data in each column so that it is easier to read.
Maximum	Display the largest value for each column of data.
Median	Display the median value for each column of data, as well as the <i>Median</i> icon.
Minimum	Display the smallest value for each column of data.
Average	Display the average value for each column of data, as well as the <i>Average</i> icon.

Interpreting Quantiles

Quantiles divide ordered data into equal-sized subsets and are the data values marking the boundaries of each subset. In BT <GO>, you can choose to organize your data into quartiles, quintiles, or deciles. Grouping data can help you identify trends

and patterns in the data.

- **Quartile:** Represents 25% of the data in a data set. The first quartile (Q1) is defined as the middle number between the smallest number and the median of the data set. The second quartile (Q2) is the median of the data. The third quartile (Q3) is the middle value between the median and the highest value of the data set. In the case of quartiles, the first quartile is the 25th percentile. This means that 25% of the data is smaller than the first quartile and 75% of the data is larger than it. In the case of the third quartile, 25% of the data is larger than it while 75% is smaller. In the case of the second quartile, which is the median, 50% of the data is smaller than it and 50% is larger than it.
- **Quintile:** Represents 20% of the data. The first quintile (Q1) represents the lowest fifth of the data (1% to 20%); the second represents the second quintile (Q2) represents the second fifth (21% to 40%); the third quintile (Q3) represents the third fifth (41% to 60%); the fourth quintile represents the fourth fifth (61% to 80%); the fifth quintile (Q5) represents the fifth fifth (81% to 100%).
- **Decile:** Represents 10% of the data. The first quintile (Q1) represents the lowest tenth of the data (1% to 10%); the second quintile (Q2) represents the second tenth fifth (11 % to 20%); the third quintile (Q3) represents the third tenth (21% to 30%); the fourth quintile represents the fourth tenth (31% to 40%); the fifth quintile (Q5) represents the fifth tenth (41% to 50%), and so on for deciles six through ten.

Strategy Alerts

You can create alerts that notify you when a rule you defined for the strategy is fulfilled in the market. Once you have created one or more alerts, you can manage them in BT <GO> or in ALRT <GO>.

For more on managing alerts: [ALRT Help Page](#).

Creating an Alert

You can create alerts to notify you when any one of the rules you defined for a strategy is fulfilled. Alerts can be set from the main library screen, the *Strategy Definition* screen, and the *Strategy Analysis* screen.

To create an alert:

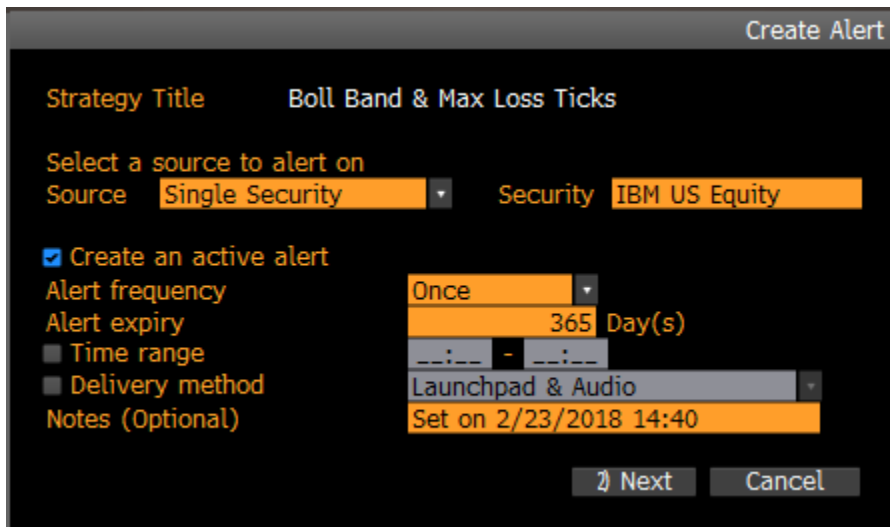
1. From the *Strategies* tab or the *Multi-Strategy* screen, select the bell icon next to a strategy.



[Hint] You can also create an alert from the *Strategy Analysis* screen. From the toolbar, select **Launch > Alert**.

The *Create Alert* window appears.

2. Follow the on-screen instructions to customize your alert (i.e., select an option from the *Source* drop-down menu).



Create Alert

Strategy Title Boll Band & Max Loss Ticks

Select a source to alert on
Source Single Security Security IBM US Equity

☒ Create an active alert

Alert frequency Once

Alert expiry 365 Day(s)

☐ Time range --:-- - --:--

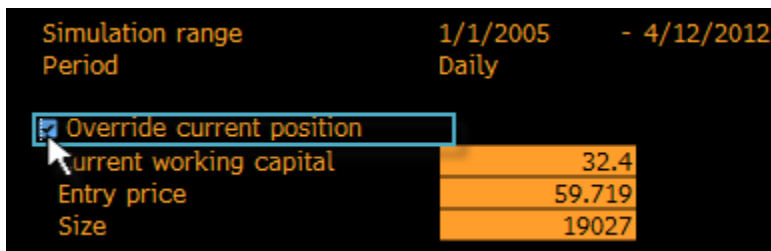
☐ Delivery method Launchpad & Audio

Notes (Optional) Set on 2/23/2018 14:40

Next Cancel

For a description of any setting, position your mouse over a label or field.

- When you have finished customizing your alert, click the **Next** button.
- If you want to edit the default position settings, click **Override current position**, then update the fields.



Simulation range 1/1/2005 - 4/12/2012

Period Daily

☒ Override current position

Current working capital 32.4

Entry price 59.719

Size 19027

- Click the **Create** button.



Create Cancel

A Success window appears. The alert icon next to the strategy is activated.

Editing an Alert

After you create an alert, you can edit it from the *Strategies* tab or the *Strategy Analysis* screen.

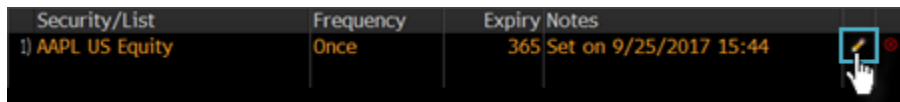
Steps:

- Click the bell icon next to a strategy with an active alert.



The *List of Alerts* window appears with a table of all the securities and lists for which this strategy has alerts in effect.

2. To edit an alert, in the *Edit* column, click the corresponding pencil icon.



The *Edit Alert* window appears with the current timing and delivery options for the alert.

3. Edit any alert settings you wish to change, then click the **Next** button.
4. Edit any position settings you wish to change, then click the **Update** button.

A *Success* window appears.

5. Continue working with your alerts, or return to backtesting:
 - To edit another alert, click the corresponding pencil icon.
 - To create a new alert, click the **Create New** button.
 - To return to the previous screen, click the **Close** button.

Simulation Controls

You can adjust a strategy's simulation controls, which affect triggers, orders, and transactions created during simulations, from the *Simulation Control* tab. This tab can be accessed when you create or review a strategy, worksheet, optimization, or multi-security analysis.

- **Date Type/Trigger Settings:** Adjust the date range and periodicity of the simulation. Choose to include today's date in the simulation and when to activate a trigger warning.
- **Order Properties:** Adjust the properties of the orders placed during the simulation, such as [Trading Approach](#) and [Initial Capital](#). For more on specific properties: [Definitions](#).
- **Transaction Properties:** Specify the properties of the transactions that occur during the simulation, such as [Reporting Currency](#) and [Commission Per Trade](#). For more on specific properties: [Definitions](#).

Reference

Calculations

Calculating Profit & Loss

The method of calculating profit and loss differs depending on the instrument type being analyzed.

- **Equities and Fixed Income**
 - Profit Long = Trade Size * (Trade Entry Price - Trade Exit Price)
 - Profit Short = - Trade Size * (Trade Entry Price - Trade Exit Price)
- **Futures**
 - Profit Long = Trade Size * Tick Value * (Trade Entry Price - Trade Exit Price)/ Tick Size
 - Profit Short = - Trade Size * Tick Value * (Trade Entry Price - Trade Exit Price)/ Tick Size

- **Currencies**

- Profit Long = Trade Size * (Trade Entry Price - Trade Exit Price)
- Profit Short = - Trade Size * (Trade Entry Price - Trade Exit Price)

Definitions

Term	Definition
%Losing ratio	1 – %Winning ratio
%Max	The maximum percent return achieved over the life of the portfolio from the start of the period.
%Max increase	The biggest increase from any low point to a high point in the portfolio value. The largest possible gain over the horizon, if the investor had bought the portfolio at the lowest point and sold at the highest.
%MaxDD	The percentage drop from an equity high to the next equity low. It reflects the change in equity as if the investor had bought the security at that high point and sold at the next low. Once a new high is reached, the percentage change from the prior high to the intervening low is recorded and a new drawdown calculation begins. Thus, several drawdowns may occur over the period of the analysis. • In the <i>Summary Statistics</i> section of the <i>Chart</i> view of the results, %Max DD measures the largest drawdown over the entire date range of the backtest. • In the <i>Trade Table</i> view of the results, %Max DD is the largest drawdown over the duration of the specific trade.
%Min	The minimum percent return achieved over the life of the portfolio from the start of the period.
%P&L	The amount of the profit and loss expressed as a percentage.
%Total	The cumulative dollar amount of loss or profit expressed as a percentage.
%Winning ratio	Gives the percent of trades that were profitable (i.e., the portfolio increased in value from one trade to the next).
Action	Allows you to specify the entry or exit trade to simulate when the factor meets one of the rules you defined for it.
All In	Allows you to specify that all the net equity will be used to calculate the trade size for each trade entry; i.e., Trade Size = Capital / Entry Price.
Analysis Title	Allows you to enter a title for a multi-security analysis.
Available Strategies	Allows you to select the strategy you want to backtest against a group of securities.
Average	The sum of the values in a data set divided by the number of values. Also known as the mean.
Avg Dur	A time (interval) statistic, the average duration is the average number of intervals in the market for each trade.
Avg Duration	A time (interval) statistic, the average duration is the average number of intervals in the market for each trade.

Term	Definition
Avg Loss	The average amount of a losing trade.
Avg P&L	The amount of the average Profit and Loss for each trade.
Avg Wins	The average amount of a profitable trade.
Bin	The interval used to group the data in the table. The bin numbers are predetermined. A data point is included in a particular bin if its number is greater than the lowest number and equal to or less than the largest number that make up the range of the data in the bin.
Capital	Allows you to specify a fixed amount of capital to be allocated for every trade in the backtest calculations; i.e., $\text{Trade Size} = \text{Capital} / \text{Entry Price}$.
Commission Per Share	Allows you to specify that the commission generated in the backtest is calculated on a per share basis.
Commission Per Trade	Allows you to specify that the commission generated in the backtest is calculated on a per transaction basis regardless of the number of shares or the total value of the transaction.
Commission Percentage	Allows you to specify that the commission generated in the backtest is calculated as a percentage of the total value of the transaction.
Condition	Allows you to specify a market move that you want to trigger an order you want the test to execute.
Contracts/Notional/Shares	Allows you to specify an explicit trade size.
Cover	Cover and Go Long and Close and Go Short actions combine the Entry and Exit actions into one rule to reverse a market position.
Crossover	The point on the chart where a security and the factor intersect. A crossover constitutes a buy or sell signal.
Currency	The currency of the investment amount used in the backtest calculations.
Daily	Allows you to select the frequency of the data for the backtest. For example, you can select data at daily or weekly intervals within the date range of the test.
Date Range	Allows you to select the date range of the data you want to backtest.
Default Trade Price	Allows you to specify the trade price you want to use for the test. The default trade price can be set to either Trade Current Close or Trade Next Open . Because the strategy is being evaluated using the Current Close, Current Close as the an entry or exit price is only valid value for execution on the Signal bar. All other evaluations of the entry or exit price are made on the bar following the Signal. This applies to next open market orders, and stop and limit orders.
Delete Column	On the <i>Optimizations</i> tab, this column is unlabeled and allows you to delete an optimization from the table. An 'X' appears in the column for each optimization.
Delta	On the <i>Market Impact</i> tab, plots the price change.
Distribution	The distribution of a data set illustrates the possible values (intervals) in the data set and how often each occurs.

Term	Definition
Enable Multiple Trades Per Bar	Allows you to permit more than one trade to appear on a single bar in a backtest result chart. When you permit more than one trade per bar, you can, for example, allow an exit to occur on the entry bar as appropriate, which is more accurate than having them appear on two bars. If left blank, the test results display only one trade per bar.
Enable Pyramiding	Enables pyramiding in the simulation and allows you to specify Scale In and Scale Out trade sizes.
End of Bar	Allows you to create an alert where a signal persists at the end of the bar. See Intrabar .
End Time	On the <i>Optimizations</i> tab, the time the optimize finished processing.
Enter Long	A buy order to take a Long position in the market.
Exit Long	A sell order to move from a Long position to a neutral position.
Expires On	On the <i>Optimizations</i> tab, the date the optimization expires. Optimizations expire 30 days after they are first submitted.
Factor	An input for a strategy. Technical studies, variables, and references to other securities or data fields can be used as inputs for a strategy.
Historical	Allows you to use historical data for a strategy when running the backtest.
Impact	The number of periods – before and after the event being analyzed – that are included in the analysis.
Initial Capital	Allows you to specify the amount of the capital investment you want to use for the test. This setting is used differently depending on the security type: • For Equities and Fixed Income, if the Trade Size field is set to <i>All in</i>, the amount in the <i>Initial Capital</i> field is used to determine initial trade size using the price as quoted. If a fixed number of shares is specified, then a check is made to ensure that sufficient initial capital is available for the first trade to occur. If you choose to trade a fixed capital amount, ensure that the amount in the <i>Initial Capital</i> field is greater than the capital amount specified in the <i>Trade Size/Capital</i> field. The capital amount available to trade will be reevaluated at each step. If losses exceed the amount available, no additional trade entries will be permitted. The initial capital is in the local currency, which means. There must be enough initial capital for at least one trade to occur. • For Futures and Currencies that trade using exchange or dealer determined explicit margining rules, the amount in the <i>Initial Capital</i> field is only used for statistical calculations such as Sharpe ratio that require a reference capital amount.
Intrabar	Allows to you set an alert that will trigger prior to the completion of a bar. This option may not result in a signal that persists at the end of the bar. See End of Bar . For example, if your signal is RSI XAbove 70, the signal may trigger early in the day, but may not be valid at the close. In this case, your alert history would show an event, but if you looked for it on a chart the next day, it would not be there.
Intraday	Allows you to use intraday data for a strategy when running the backtest .

Term	Definition
Long	The number of long trades or the cumulative dollar amount of loss or profit for all long trades. Long trading involves buying futures contracts in the hope that the price will increase. Long trading is defined as holding an investment for an extended period of time.
Losses	The number of trades that lost money.
Market Order	Close of the signal bar or trigger bar. Previously this was the Current Close.
Max	The upper bound in a range of values to be optimized for a parameter.
Max Adv Ex	A risk indicator, Maximum Adverse Excursion is the highest loss incurred for the trade before it closes. This value is considered useful for identifying trades that offer a lower-than-average profit potential.
Max DD length	The length of time from the start of %Max DD until the low point is reached.
Max DD Recovery Period	The sum of Max DD length and Recovery from Max DD
Max Fav Ex	A risk indicator, Maximum Favorable Excursion is the maximum profit achieved for the trade before it closes. This value is considered useful for identifying trades that offer a higher-than-average profit potential.
Max Increase	The length of time from the start of %Max increase until the high point is reached.
Max Loss	The largest loss realized by a trade.
Max Wins	The largest profit realized by a trade.
Median	The middle value in a data set where half of all the values in the data are larger than the median and half are smaller.
Min	The lower bound in a range of values to be optimized for a parameter.
Minutes before bar's end	Allows you to specify the time (in minutes) before the end of the test when, a condition defined for any of the strategy rules has been achieved in the test, you want to be alerted. The alert is delivered according to the settings in the <i>Edit Strategy Alerts</i> window. For more information on alerts, see Strategy Alerts .
Name	The specific source of the securities that you want to backtest.
Next Open	The opening price at the open of the next day's market.
Notes	Allows you to enter notes about a multi-security analysis.
Notional Amount	The face value of a financial instrument.
NumBars	A time (interval) statistic, NumBars is the number of bars (intervals) used to calculate the average duration.
Offset	Moves the point of analysis to a different position before or after the event.
Optimization Title	The title of a strategy you have optimized.
Order Properties	Allows you to define how an order will be constructed when a signal to buy or sell is triggered.
Order Type	Allows you to specify the type of order you want the backtest to execute under a specific market condition.

Term	Definition
P/L	The profit or loss expressed as a percentage.
P&L	Allows you to show or hide a profit and loss chart below the main chart that plots the results of the backtest.
Periodicity	The interval applied to the analysis (e.g., daily, monthly)
Price Slippage	Adjust the difference between the expected price and the actual price when a trade is executed in the simulation.
Profit Factor	The net profit of winning trades/net loss of losing trades.
Qualifier	Allows you to designate the function of each element of a rule you define in a strategy.
Quantile	A value that divides a set of data into equal-sized groups.
Quartiles	A statistical term that describes the division of data observations into four intervals based on data value and how the values in each interval compare to all of the observations.
Quintiles	A statistical term that describes the division of data observations into five intervals based on data value and how the values in each interval compare to all of the observations.
Recovery from Max DD	The time from the low point until the previous high is reach again.
Reporting Currency	Allows you to report in a currency other than the local currency of the security being backtested.
Return	On the <i>Market Impact</i> tab, this option plots the percent change.
Security List	Allows you to activate the <i>Source</i> and <i>Name</i> fields so you can specify the group of securities that you want to backtest, e.g., equity index, portfolio, or other source.
Security/Datafield	A factor that allows you to add a data element (for example, volume) for the parent security, or a different security, to a strategy.
Selected Strategies	The strategy you want to backtest against a group of securities.
Sharpe Ratio	A risk indicator, the Sharpe Ratio is a measure of the excess return (risk premium) per unit of risk. The Sharpe Ratio indicates the return versus the risk of an investment. A higher Sharpe Ratio number indicates a greater return. This ratio is not applicable for intraday periodicities.
Short	The number of short trades or the cumulative dollar amount of loss or profit for all short trades. Short-term trading is defined as holding an investment for a brief period of time in the hope of making quick profit.
Slippage	Allows you to specify the price difference, in units per share, between the signal price and the actual trade execution price to apply. The slippage is applied to the trade price by being added to the buy signal trade price and subtracted from the sell signal trade price.

Term	Definition
Sortino	A ratio developed by Frank A. Sortino differentiates good from bad volatility in the Sharpe Ratio. The differentiation allows a risk-adjusted measure of the security's performance without penalizing it for upward price changes by using a downside rather than standard deviation for the denominator. The Sharpe Ratio uses the standard deviation. Note: This ratio is not applicable for intraday periodicities.
Source	The source of the securities for which you are running a backtest.
Start Time	On the <i>Optimizations</i> tab, the time the optimization began processing.
Status	On the <i>Optimizations</i> tab, the current status of the optimization. The options are: - Submitted on: The date you submitted the strategy for optimization. - Start Time: The time the optimization began processing. - End Time: The time the optimize finished processing.
Step	The interval to be used between optimization runs.
Today	Allows you start the backtest with the date and time that the simulation is submitted.
Total Loss	The total amount of losing trades.
Total P&L	The total number of trades or the cumulative dollar amount of the loss or profit.
Total Trades	The total number of trades.
Total Wins	The total amount of profitable trades.
Trade Current Close	The most recent closing price.
Trade Next Open	The next day's opening price. The next day refers to the next day, week or month, depending on the frequency of the data in the backtest.
Trade Price	Allows you to specify the type of price at which the buy/sell trade is executed: - Trade Current Close: The most recent closing price. - Trade Next Open: The next day's opening price. The next day refers to the next day, week or month, depending on the frequency of the data in the backtest. The slippage amount is added to the trade price in this field.

Term	Definition
Trade Size	- For equities and fixed income instruments: - All In: Allows you to specify that all the net equity will be used to calculate the trade size for each trade entry; i.e., $\text{Trade Size} = \text{Capital} / \text{Entry Price}$. - Capital: Allows you to specify the amount of capital to be allocated for every trade in the backtest calculations; i.e., $\text{Trade Size} = \text{Capital} / \text{Entry Price}$. - Contracts/Notional/Shares: Allows you to specify an explicit trade size. - For futures: Specified as the fixed number of contracts that will be used for each trade. - For currencies: Specified as a fixed notional amount for the currency being traded. On the <i>Factors</i> side panel, allows you to show or hide a trade size chart below the main chart that plots the results of the backtest.
Trades	The number of trades.
Trading Approach	Allows you to specify the type of trade(s) you want the strategy to test. If the strategy contains rules that conflict with the trading approach chosen, they will be removed or adapted when you save the settings. You can choose: Long & Short: Include long and short trades in the backtest. This is the default trading approach Long: Include only long trades in the backtest. - When switching from a Long & Short strategy to a Long Only strategy, if the strategy contained the reversal rule Cover and Go Long it will be converted to an Enter Long rule. If the strategy contained a Close and Go Short rule, it will be converted to an Exit Long rule. Short: Include only short trades in the backtest. - When switching from a Long & Short strategy to a Short Only strategy, if the strategy contained the reversal rule Close and Go Short it will be converted to an Enter Short rule. If the strategy contained a Cover and Go Long rule, it will be converted to an Exit Short rule. The entry in this field determines the actions that are available when setting up strategy rules, e.g., choosing Long Only results in only long options appearing in the Action drop-down in the <i>Rules</i> section of the <i>Strategy Definition</i> tab.

Term	Definition
Trigger Warning	Allows you to set a trigger to alert you when any one of the rules you defined for a strategy is fulfilled. The alert is delivered via the delivery method specified in the <i>Edit Strategy Alerts</i> window. For more information on alert properties, see Strategy Alerts .
User Defined Studies	A user-defined technical study that you can add as a factor to a strategy.
Variable	A numeric field that is used in defining a rule in place of a specific value.
Wins	The number of trades for which a profit was made.

Take the next step.

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